

EMPLOYEE MANAGEMENT SYSTEM

Software Requirements Specification

Product	Employee Management System (EMS) — Pixous HR Portal
Document type	Software Requirements Specification (SRS)
Version	1.0
Status	Baseline — derived from the implemented source code
Date	30 July 2026

This document specifies the complete functional and non-functional requirements of the Employee Management System: the actors it serves, the data it holds, the rules it enforces, the interfaces it exposes, and the quality attributes it must satisfy.

Table of Contents

If the entries below are not yet filled in, select all (Ctrl+A) and press F9, then choose "Update entire table".

Table of Contents.....	2
Part I — Introduction and Overview.....	7
1. Introduction.....	7
1.1 Purpose.....	7
1.2 Product Scope.....	7
1.3 Definitions, Acronyms and Abbreviations.....	7
1.4 Document Conventions and Requirement Identifiers.....	8
1.5 Intended Audience and Reading Order.....	9
1.6 References.....	9
2. Overall Description.....	9
2.1 Product Perspective.....	9
2.2 System Context.....	10
2.3 Product Functions Summary.....	10
2.4 Operating Environment.....	11
2.5 Design and Implementation Constraints.....	12
2.6 Assumptions and Dependencies.....	13
2.7 User Documentation.....	13
3. External Interface Requirements.....	13
3.1 User Interface Requirements.....	13
3.2 Hardware Interface Requirements.....	14
3.3 Software Interface Requirements.....	15
3.4 Communication Interface Requirements.....	15
3.5 API Conventions.....	16
4. Users, Roles and Access Control.....	16
4.1 User Classes.....	16
4.2 Role Catalogue.....	17
4.3 Permission Catalogue.....	18
4.4 Role to Permission Matrix.....	18
4.5 Endpoint Authorisation Rules.....	19
4.6 Approval Routing Hierarchy.....	20
Part II — Functional Requirements.....	22
5. Authentication and Session Management.....	22
5.1 Sign-in.....	22
5.2 Account Lockout.....	22
5.3 Tokens and Session Lifetime.....	22
5.4 Self-service Registration.....	22

5.5 Password Management	23
5.6 Availability Checks	23
5.7 Current User	23
6. Employee Management.....	23
6.1 Administrative Employee Creation	23
6.2 Bulk Employee Creation.....	24
6.3 Employee Directory	24
6.4 Profile Read and Maintenance	24
6.5 Credentials Administration	25
6.6 Photograph and Documents.....	25
6.7 Bank Details	25
6.8 Teams.....	25
6.9 Offboarding and Deletion	25
7. Organisation and Master Data.....	26
7.1 Reference Lists	26
7.2 Teams as Designations.....	26
7.3 Locations, Sites and Shifts.....	26
8. Attendance Management	26
8.1 Punch In	26
8.2 Geofence Evaluation.....	27
8.3 Lateness	27
8.4 Punch Out	27
8.5 Views and Summaries	27
8.6 Face-based Verification	28
9. Leave Management	28
9.1 Leave Types and Policy	28
9.2 Balances and Allocation	28
9.3 Working-day Calculation.....	29
9.4 Applying for Leave	29
9.5 Approval Notification.....	29
9.6 Own Requests and Cancellation	29
9.7 Approver Queues.....	30
9.8 Organisation Leave Calendar	30
9.9 Decisions	30
9.10 Loss-of-Pay Preview	30
10. Hourly Permission Requests	31
11. Payroll and Payslips.....	31
11.1 Salary Structures	31
11.2 Direct Payslip Generation	32

11.3 Payslip Request Workflow	32
11.4 Payslip Access	33
11.5 Payslip Document	33
11.6 Payroll Runs	33
12. Task Management	34
12.1 Assignment	34
12.2 Progress and Completion	34
12.3 Editing and Removal	35
12.4 Views and Export	35
13. Work Reports	35
14. Travel and Expense Claims.....	36
14.1 Travel Claims	36
14.2 General Expense Claims.....	36
15. Asset Management	37
16. Support Helpdesk.....	37
17. Complaints and Needs	39
18. Safety Incident Management.....	39
19. Onboarding and Offboarding.....	40
20. Performance Management.....	40
21. Dashboards and Analytics.....	40
21.1 Personal Dashboard	40
21.2 Executive Dashboard	41
21.3 Celebrations	41
21.4 Side-service Analytics.....	41
22. Notifications and Alerts	41
23. Internal Communication	42
23.1 Channels	42
23.2 Messages	42
24. AI Assistant	43
25. Reports and Data Export.....	43
26. Calendar and Holidays	43
27. File Storage and Documents	44
28. System Settings.....	44
29. Mobile Application.....	44
30. Biometric and Document Intelligence Service	44
Part III — Data, Rules and Quality	46
31. Data Requirements	46
31.1 Logical Entity Inventory	46
31.2 Common Column Conventions	46

31.3 Key Entity Definitions.....	46
31.4 Reference Data and Seeds	48
31.5 Audit, Retention and Integrity	48
31.6 Schema Evolution	48
32. Consolidated Business Rules	49
Calendar and time	49
Attendance	49
Leave.....	49
Hourly permissions	50
Payroll	50
Tasks	51
Claims.....	51
Assets.....	52
Helpdesk	52
Complaints, safety and general	52
33. Non-Functional Requirements.....	53
33.1 Performance	53
33.2 Scalability and Capacity	53
33.3 Security	53
33.4 Reliability and Availability	54
33.5 Usability and Accessibility.....	54
33.6 Compatibility and Portability	54
33.7 Maintainability.....	55
33.8 Localisation and Internationalisation.....	55
33.9 Observability	55
33.10 Backup and Recovery.....	55
33.11 Compliance and Privacy	55
34. Deployment and Operations.....	56
34.1 Environments.....	56
34.2 Configuration	56
34.3 Continuous Deployment	56
34.4 Operations	57
35. Constraints, Assumptions and Known Limitations	57
35.1 Known Limitations	57
35.2 Explicit Non-Goals.....	58
36. Traceability Matrix.....	58
37. Appendices	60
Appendix A — Status Enumerations	60
Appendix B — State Machines.....	60

Appendix C — Generated Identifier Formats	61
Appendix D — Configuration Defaults.....	61
Appendix E — Notification Catalogue.....	62
Appendix F — SMS Catalogue.....	63
Appendix G — Requirement Count Summary	63
Appendix H — Document Revision History	63

Part I — Introduction and Overview

1. Introduction

1.1 Purpose

This document specifies the complete functional and non-functional requirements of the **Employee Management System (EMS)** — a multi-role, multi-division human-resources platform that manages the full employee lifecycle from joining to exit.

The specification describes **what the system does and must continue to do**: the actors it serves, the data it holds, the rules it enforces, the interfaces it exposes, and the quality attributes it must satisfy. It is written to be authoritative for development, maintenance, enhancement, contract acceptance, and operational hand-over.

1.2 Product Scope

The EMS is a single platform serving an organisation that operates two business divisions:

Division	Internal code	Displayed as	Workforce character
Information Technology / Digital	IT	Digital	Desk-based staff at fixed office locations
Civil / Infrastructure	CIVIL	Infra	Field and site-based staff across project sites

Both divisions share one deployment, one identity store, one permission model and one database, with division-aware behaviour where the two differ (site-based versus office-based attendance, machinery versus IT assets, site safety reporting).

In scope. Identity and access control; employee records and documents; organisation master data; GPS-validated attendance; leave entitlements, applications and multi-level approvals; hourly permission requests; salary structures, payslip generation, payslip request workflow and payroll runs; task assignment and progress tracking; daily work reporting; travel/expense claims; asset registry, allocation and QR tagging; a support helpdesk with SLA and satisfaction rating; complaints and needs; safety incidents; onboarding checklists and offboarding records; performance goals and reviews; role-aware dashboards; real-time notifications with SMS fan-out; internal group and direct messaging; a multilingual AI assistant; spreadsheet exports; the holiday calendar; system settings; a companion mobile application; and a biometric/document-intelligence side service.

Out of scope. Recruitment and applicant tracking; a learning-management system; statutory e-filing and tax return submission; general ledger and accounts integration; multi-tenant or multi-company isolation; biometric hardware device integration; and full offline operation.

1.3 Definitions, Acronyms and Abbreviations

Term	Meaning
EMS	Employee Management System — this product
RBAC	Role-Based Access Control
JWT	JSON Web Token — the bearer credential for API calls
Access token	Short-lived signed token proving identity on each request
Refresh token	Opaque, database-backed token exchanged for a new access token
Permission	A named, atomic capability (e.g. LEAVE_APPROVE) checked at the endpoint

Term	Meaning
Role	A named bundle of permissions assigned to a user (e.g. IT_HR)
Team	A group of employees sharing the same designation title
Designation title	Free-text team/job name carried on the employee record; the primary team key
TL	Team Leader (role IT_TL)
Geofence	A circular boundary (centre coordinate + radius in metres) around an office or site
Punch	A single attendance event — punch-in or punch-out
WFH	Work From Home attendance mode
LOP	Loss of Pay — an unpaid absence day that reduces net pay
Permission request	An hours-wise short absence during a working day (distinct from an RBAC permission)
TA claim	Travel Allowance / travel-and-conveyance expense claim
SLA	Service Level Agreement — the due-by time for a support ticket
PF / ESI / PT / TDS	Provident Fund / Employees' State Insurance / Professional Tax / Tax Deducted at Source
Payslip request	An employee-initiated request that results in an admin-generated payslip
Payroll run	A batch generation of payslips for one month, moving through a status workflow
Announcement channel	A broadcast messaging group every employee automatically belongs to
Direct room	A hidden two-member messaging group backing a one-to-one conversation
STOMP	Simple Text Oriented Messaging Protocol, used over WebSocket for live push
Master data	Reference lists (departments, designations, shifts, sites, blood groups, ...)
Vault copy	A reversibly-encrypted copy of a password, readable by HR and administrators

1.4 Document Conventions and Requirement Identifiers

- "shall" denotes a mandatory requirement.
- "should" denotes a recommended requirement.
- "may" denotes an optional or permitted behaviour.
- Identifiers are stable and must not be renumbered; superseded requirements are marked, never deleted.

Prefix	Category	Example
FR-<MOD>-nnn	Functional requirement, per module	FR-ATT-004
NFR-<CAT>-nnn	Non-functional requirement	NFR-SEC-003
BR-nnn	Business rule	BR-021
DR-nnn	Data requirement	DR-007

Prefix	Category	Example
IR-nnn	Interface requirement	IR-005
OR-nnn	Operational requirement	OR-004
CN-nnn	Constraint	CN-002
AS-nnn	Assumption	AS-003
LM-nnn	Known limitation	LM-005

Module codes used in functional identifiers:

AUTH authentication · EMP employee management · ORG organisation master data · ATT attendance · LV leave · PRM hourly permissions · PAY payroll · TSK tasks · WRP work reports · CLM travel & expense claims · AST assets · HD helpdesk · CMP complaints & needs · SAF safety · ONB onboarding & offboarding · PRF performance · DSH dashboards · NOT notifications · CHT communication · BOT AI assistant · RPT reports · CAL calendar · FIL files · SET settings · MOB mobile · BIO biometric service

1.5 Intended Audience and Reading Order

Audience	Suggested reading
Business sponsors, HR leadership	§1, §2, §4, §5–§30 headings, §33
Solution architects	§2, §3, §31, §33, §34
Developers	§3 onwards in full, especially §32 and §37
Operations and support	§3.3–§3.5, §28, §33, §34, §37
Compliance and audit	§4, §31.5, §32, §33.3
Acceptance reviewers	§5–§32, §36

1.6 References

Ref	Document / artefact
R1	Application source repository (backend, web, mobile, analytics service)
R2	Database migration set — the authoritative schema definition
R3	OpenAPI / Swagger UI published by the running API at /swagger-ui.html
R4	Architecture notes and setup guide under docs/
R5	Deployment guide, container composition and deployment scripts
R6	Requirements analysis and traceability notes under docs/

2. Overall Description

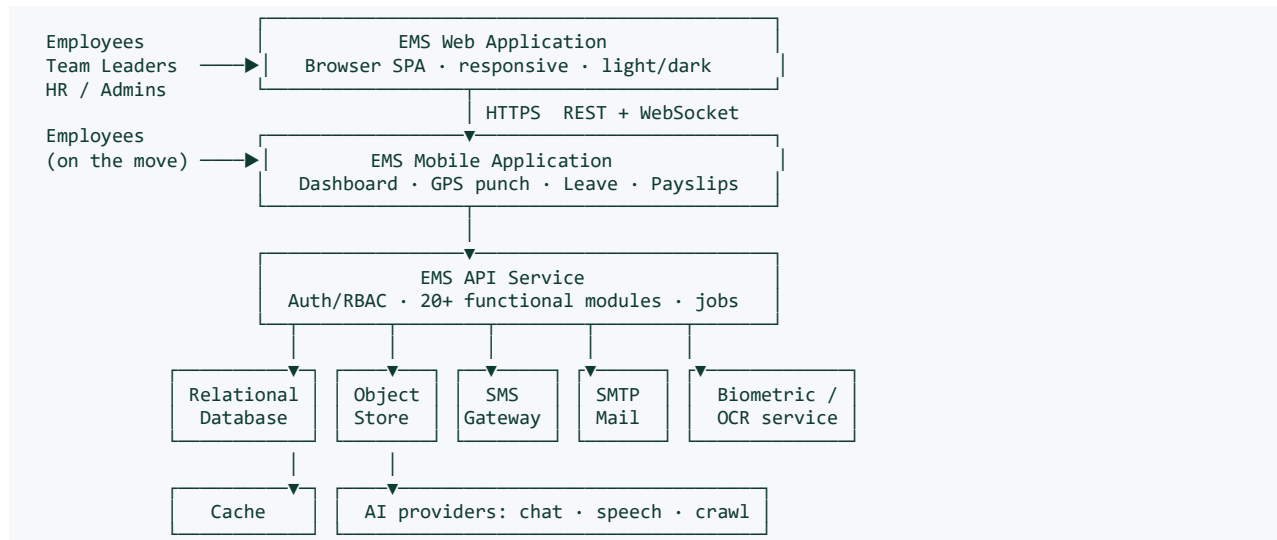
2.1 Product Perspective

The EMS replaces a legacy PHP-based employee API and consolidates several manual, spreadsheet-driven HR processes into one system of record. It deliberately preserves the shape of a number of legacy contracts so that existing integrations and habits carry over:

- master data is served by a single **dropdown-by-type** endpoint;
- bank details follow an **add / update / delete / view** shape;
- payslips are composed of the same **basic / HRA / allowances / deductions** components;
- profile and photo updates map one-to-one onto the legacy self-service calls.

On top of that foundation the system layers attendance, leave, tasks, claims, assets, helpdesk, complaints, safety, messaging and dashboards.

2.2 System Context



IR-001 The system shall present exactly one public network entry point in production; all internal services shall be unreachable from outside the deployment network.

IR-002 The web application, the mobile application and any future client shall consume the same versioned REST API and the same response envelope.

2.3 Product Functions Summary

#	Capability	Primary actors
1	Username/password sign-in, token refresh, self password change, account lockout	All users
2	Employee creation (single and bulk), profile maintenance, credential reset, document storage, deletion	HR, Admin
3	Master data: departments, designations/teams, positions, employment statuses, blood groups, office locations, shifts, sites, holidays	HR, Admin
4	GPS/geofence attendance punch, WFH and site modes, lateness and overtime, monthly summary, team and organisation views, absentee list	All users
5	Leave types and policies, annual balance allocation, application with quota/notice/gender/quarterly checks, single-level approval routing, cancellation, calendar	All users

#	Capability	Primary actors
6	Hourly permission requests with single-level routing and overdue protection	All users
7	Salary structures, statutory computation, PDF payslips, payslip request/approve workflow, monthly payroll runs with finance approval	Admin, HR, Finance
8	Task assignment with scope rules, priority, forward-only progress, export	HR, Team Leaders, Employees
9	Daily work reports with team and organisation roll-ups and export	All users
10	Travel/expense claims with kilometre and fare breakdown, attachments, approval	All users, HR
11	Asset registry with quantity, QR tags, allocation, acknowledgement and return	Asset managers, HR
12	Support tickets with SLA, ordered status workflow, threaded comments, rating	All users, HR agents
13	Complaints and needs with reference codes and HR response	All users, HR
14	Safety incident reporting with optional anonymity and investigation workflow	All users, safety staff
15	Onboarding checklists and offboarding records	HR, Admin
16	Performance goals and appraisal reviews	Employees, Managers
17	Personal dashboard, executive dashboard, celebrations panel	All users, Executives
18	Persistent in-app notifications, live push, SMS fan-out, scheduled celebration alerts	System
19	Group channels, announcement broadcast, direct messages, team rooms, voice notes, attachments, call signalling	All users
20	Multilingual AI assistant with knowledge ingestion, translation, speech synthesis and transcription	All users, Admin
21	Attendance, leave and payroll spreadsheet exports	Report viewers
22	Holiday calendar with organisation-wide notification	HR, Admin
23	Configurable system settings and provider credentials	Admin

2.4 Operating Environment

Server side

Element	Requirement
Runtime	Java 17 or later, Spring Boot 3.5.x application server
Database	MySQL 8.4 (InnoDB, utf8mb4)
Cache	Redis (optional at runtime; absence shall not affect health)
Event streaming	Kafka (optional; used for chat event distribution)
Mail	SMTP relay (optional at runtime)
Side service	Python 3 FastAPI service for face matching and OCR (optional profile)
Reverse proxy	Nginx terminating public traffic and routing to internal services
Container platform	Docker with Compose orchestration on a single host
Server time zone	Asia/Kolkata for all containers that record or render timestamps

Client side

Element	Requirement
Web	Current versions of Chrome, Edge, Firefox and Safari
Web features used	Fetch/XHR, WebSocket, localStorage, Geolocation, MediaDevices (camera/microphone), Web Speech synthesis
Viewport	Usable from 320 px width upwards through desktop
Mobile	Android and iOS via a React Native (Expo) application
Mobile features used	Secure credential storage, foreground location

2.5 Design and Implementation Constraints

CN-001 The relational schema shall be owned exclusively by versioned forward-only migrations; the ORM shall validate against the schema and shall never create or alter it.

CN-002 Every migration shall be idempotent and safe to apply to both an empty database and a populated production database.

CN-003 All API responses shall use a single uniform envelope, and all paginated responses a single uniform page wrapper.

CN-004 Authorisation shall be expressed declaratively at the endpoint using permission or role expressions, never by ad-hoc checks scattered through presentation code.

CN-005 The API shall be stateless; no server-side HTTP session shall be created.

CN-006 No secret — signing key, database password, SMS key, AI provider key — shall be committed to source control; all shall be supplied by environment variables or database-held settings.

CN-007 The token signing secret shall have no fallback default outside the development profile; the application shall refuse to start without it.

CN-008 Configurable operational values (geofence radius, office hours, lateness grace, lock thresholds, SLA windows, accrual caps) shall be held in configuration or master data, never hard-coded in business logic.

CN-009 Money shall be represented and computed as fixed-point decimal values scaled to two places with half-up rounding; floating-point arithmetic shall not be used for monetary results.

CN-010 Geographic coordinates shall be stored with seven decimal places of precision.

CN-011 Uploaded content shall never be executed or interpreted; stored files shall be served as opaque byte streams.

2.6 Assumptions and Dependencies

AS-001 The organisation observes a single weekly off on **Sunday; Saturday is a working day**. **AS-002** Standard office hours are **09:00 to 18:00** local time unless a shift overrides the start. **AS-003** Currency is Indian Rupees; statutory deductions follow Indian norms. **AS-004** Mobile numbers are Indian ten-digit numbers with country code +91. **AS-005** Employees possess a device capable of supplying location for field attendance; absence of location shall degrade gracefully rather than block a punch. **AS-006** SMS delivery is best-effort and depends on an external gateway; failure shall never block the originating business action. **AS-007** AI assistant capability is optional and depends on externally supplied provider credentials. **AS-008** Only one organisation is served per deployment.

2.7 User Documentation

FR-DOC-001 The system shall publish live, interactive API documentation for every endpoint, grouped by functional area, with request and response schemas. **FR-DOC-002** The repository shall carry a setup guide, an architecture description, a deployment guide and this specification. **FR-DOC-003** Screens that accept bulk uploads shall display the exact expected workbook layout and shall offer a downloadable blank template that matches the parser.

3. External Interface Requirements

3.1 User Interface Requirements

FR-UI-001 The web application shall present a persistent left navigation sidebar, a top bar carrying the notification bell, theme toggle and account menu, and a routed content area.

FR-UI-002 Navigation entries shall be filtered by the signed-in user's permissions and roles. An entry the user cannot use shall not be rendered.

FR-UI-003 Navigation shall support collapsible groups. Leave Management and Payroll shall be groups, expanded automatically when the current route falls inside them.

FR-UI-004 The following navigation structure shall be provided:

Entry	Route	Visibility
Dashboard	/	All
Employees	/employees	USER_MANAGE or ATTENDANCE_TEAM or REPORT_VIEW; hidden from Team Leaders
Attendance	/attendance	All except Super Admin
Employee Attendance / Team Attendance	/team-attendance	ATTENDANCE_TEAM; labelled "Team Attendance" for Team Leaders, "Employee Attendance" otherwise
Leave Management → Leave	/leave	All except Super Admin
Leave Management → Permission	/leave/permissions	All
Leave Management → Approvals	/leave/approvals	LEAVE_APPROVE
Leave Management → Leave Policies	/leave/policies	ORG_MANAGE
Payroll → Payroll	/payroll/requests	PAYROLL_RUN
Payroll → Payslips	/payslips	All except Super Admin, HR roles
Payroll Runs	/payroll/run	PAYROLL_RUN or PAYROLL_APPROVE
Work Reports	/work-reports	All

Entry	Route	Visibility
Tasks	/tasks	All
Teams	/teams	All — all-teams view for USER_MANAGE/IT_MGR, own-team view otherwise
Documents	/documents	ORG_MANAGE; hidden from Super Admin
Calendar	/calendar	All
Claims	/ta-expenses	All
Assets	/assets	All
Supports	/helpdesk	All
Complaints	/complaints	All
Reports	/reports	REPORT_VIEW; hidden from Super Admin
Communities	/communities	ORG_MANAGE
Chat	/chat	All
Onboarding	/onboarding	USER_MANAGE
AI Assistant Settings	/admin/ai-assistant	USER_MANAGE
Notifications	/notifications	All
Profile	/profile	All

FR-UI-005 Unauthenticated access to any protected route shall redirect to the sign-in screen. An authenticated user reaching a route their permissions exclude shall be shown an access-denied result rather than a blank screen.

FR-UI-006 The interface shall offer light and dark themes, shall remember the chosen theme across sessions, and shall render legibly in both.

FR-UI-007 Every data grid shall support pagination, and where the underlying endpoint supports it, text search and filtering.

FR-UI-008 Every asynchronous action shall present a loading state, and shall present success and failure feedback as a transient notification carrying the server's message.

FR-UI-009 Role labels displayed on screen shall be humanised. Specifically, the stored code IT_MGR shall be displayed as IT_HR wherever a raw role code is shown, and the account role shown beside the user's name shall use the friendly names: Board Admin, Super Admin, HR Head, HR, Team Leader, Employee, Site Supervisor, Field Employee.

FR-UI-010 The notification bell shall show an unread count, shall list recent notifications with a per-type icon and colour, shall allow marking one or all as read, and shall navigate to the notification's target route when selected.

FR-UI-011 An AI assistant launcher shall be available from every authenticated screen.

FR-UI-012 Empty result sets shall render an explanatory empty state rather than an empty table.

FR-UI-013 Screens presenting images shall offer a full-screen lightbox view.

3.2 Hardware Interface Requirements

IR-003 The client shall request device geolocation for attendance punches. If permission is refused or unavailable, the punch shall still be accepted and shall be recorded as outside the geofence with an exception flag.

IR-004 The client shall be able to capture still images from a device camera for face-based verification and document capture, and audio from a device microphone for voice messages and speech input.

IR-005 Generated asset QR codes shall be renderable at a size suitable for physical label printing and scannable by commodity scanners.

3.3 Software Interface Requirements

Ref	Interface	Direction	Purpose	Failure behaviour
IR-006	MySQL 8.4	Outbound	System of record	Fatal — application unhealthy
IR-007	Redis	Outbound	Cache	Non-fatal; excluded from health
IR-008	Kafka	Outbound/inbound	Chat event publication and consumption	Non-fatal; direct delivery continues
IR-009	SMTP	Outbound	E-mail delivery	Non-fatal; excluded from health
IR-010	Primary SMS gateway	Outbound	Transactional SMS on the quick route for ten-digit Indian numbers	Logged and skipped
IR-011	Secondary SMS gateway	Outbound	Fallback SMS to E.164 numbers	Logged and skipped
IR-012	LLM chat providers	Outbound	AI assistant conversation	Assistant returns a graceful message
IR-013	Speech-to-text provider	Outbound	Voice input transcription	Feature reported unavailable
IR-014	Text-to-speech provider	Outbound	Spoken assistant replies	Feature reported unavailable
IR-015	Web crawl provider	Outbound	Ingest a website into the assistant knowledge base	Ingestion fails with a message
IR-016	Biometric/OCR service	Outbound	Face enrolment and verification, numeric OCR, analytics	Feature reported unavailable
IR-017	Object/file storage	Outbound	Documents, photos, payslip PDFs, QR images, attachments	Fatal for the affected operation only

IR-018 SMS dispatch shall be asynchronous and shall never propagate an exception into the calling business transaction.

IR-019 The primary SMS gateway shall be used whenever it is enabled and holds a key; otherwise the secondary gateway shall be used; if neither is configured the attempt shall be logged and skipped.

IR-020 AI provider credentials shall be stored server-side and shall never be returned to a client in full; any display shall be masked to a short prefix and suffix.

3.4 Communication Interface Requirements

IR-021 All client–server traffic shall use HTTP over TLS in production.

IR-022 The API shall be rooted at `/api` and shall accept and return JSON, except for file downloads and uploads.

IR-023 Cross-origin access shall be restricted to an explicitly configured list of allowed web origins.

IR-024 A STOMP-over-WebSocket endpoint shall be exposed at `/ws` with SockJS fallback, brokering `/topic` and `/queue` destinations, accepting client sends under `/app` and resolving user destinations under `/user`.

IR-025 The server shall push each notification to both the recipient's user-scoped queue and a per-user public topic, so that clients subscribing either way receive it.

IR-026 File uploads shall accept up to **25 MB per file** and **60 MB per request**; the reverse proxy shall permit the same.

3.5 API Conventions

IR-027 Every response shall be wrapped as:

```
{
  "success": true,
  "message": "OK",
  "data": { },
  "errors": null,
  "timestamp": "2026-07-30T10:15:30Z"
}
```

Null members shall be omitted from serialisation.

IR-028 Every paginated response shall carry `content`, `page`, `size`, `totalElements`, `totalPages` and `last`.

IR-029 Errors shall be mapped from canonical error codes to HTTP statuses:

Error code	HTTP status
VALIDATION_ERROR	400 Bad Request
BAD_CREDENTIALS	401 Unauthorized
UNAUTHENTICATED	401 Unauthorized
TOKEN_EXPIRED	401 Unauthorized
ACCOUNT_LOCKED	423 Locked
ACCESS_DENIED	403 Forbidden
NOT_FOUND	404 Not Found
CONFLICT	409 Conflict
GEOFENCE_VIOLATION	422 Unprocessable Entity
BUSINESS_RULE	422 Unprocessable Entity
INTERNAL	500 Internal Server Error

IR-030 A request with no credentials or an expired token shall receive **401** with a message directing the user to sign in again. A request from an authenticated user lacking the required permission shall receive **403**. The two shall never be conflated, because clients treat 401 as "refresh and retry" and 403 as "not allowed".

IR-031 Field validation failures shall return 400 with a map of field name to message.

IR-032 Unhandled exceptions shall return 500 with a generic message; internal detail shall be logged, not returned.

IR-033 The following paths shall be publicly reachable without a token: the root path, all `/api/auth/**` endpoints, `/api/files/**`, the API documentation paths, the WebSocket handshake, the health endpoint and the error path. Every other path shall require authentication.

IR-034 A health endpoint shall be exposed at `/actuator/health` reporting database reachability, without detail disclosure, and excluding optional cache and mail indicators.

4. Users, Roles and Access Control

4.1 User Classes

Class	Roles	Characteristics
-------	-------	-----------------

Class	Roles	Characteristics
Employee	IT_EMP, CV_EMP	Self-service only: own attendance, leave, permissions, payslips, tasks, work reports, claims, assets, tickets, complaints, chat
Team Leader	IT_TL	Employee capabilities plus own-team attendance, short-leave and small-leave approval, task assignment within team, team claim and work-report visibility
HR	IT_MGR, IT_HR, CV_HR	Employee lifecycle, teams, leave and claim decisions, payroll requests, helpdesk agency, complaints, assets, calendar, task assignment to leaders, organisation-wide visibility
Supervisor	CV_SUP	Site leave approval, site attendance, site reporting
Finance	IT_FIN	Payroll run approval and financial reporting
Executive	IT_CEO	Executive dashboard and reports
Asset Manager	IT_AST, CV_AST	Asset lifecycle and helpdesk agency
Facilities Admin	CV_ADM	Sites, facilities, assets, helpdesk agency, master data
Administrator	SUPER_ADMIN, BOARD_ADMIN	Full configuration and organisation-wide access

4.2 Role Catalogue

FR-RBAC-001 The system shall provide the following roles, each scoped to a division:

Code	Name	Division
SUPER_ADMIN	Platform Super Admin	Both
BOARD_ADMIN	Board Admin (equivalent access to Super Admin)	Both
IT_EMP	IT Employee	IT
IT_TL	Team Leader	IT
IT_MGR	IT Manager / Team Lead — displayed as HR	IT
IT_HR	IT HR / Payroll Manager — displayed as HR Head	IT
IT_FIN	Finance Officer	IT
IT_CEO	CEO	IT
IT_AST	IT Asset Manager	IT
CV_EMP	Civil Site Employee	Civil
CV_SUP	Civil Supervisor	Civil
CV_HR	Civil HR Manager	Civil
CV_ADM	Civil / Facilities Admin	Civil

Code	Name	Division
CV_AST	Civil Asset Manager	Civil

4.3 Permission Catalogue

FR-RBAC-002 The system shall provide the following permissions:

Code	Capability granted
USER_MANAGE	Full employee administration, including deletion, offboarding and team detachment
EMPLOYEE_MANAGE	Create and edit employee accounts and their bank details and documents
TEAM_MANAGE	Create and delete teams
ORG_MANAGE	Maintain organisation master data and leave policies
CALENDAR_MANAGE	Add and remove calendar entries
LEAVE_APPLY	Apply for leave
LEAVE_APPROVE	Decide leave and hourly permission requests
ATTENDANCE_SELF	Record own attendance
ATTENDANCE_TEAM	View team and organisation attendance
PAYROLL_VIEW	View payslips and salary structures
PAYROLL_RUN	Maintain salary structures, generate payslips, run payroll, decide payslip requests
PAYROLL_APPROVE	Give finance approval to a payroll run
CLAIM_APPROVE	Approve or reject expense claims
ASSET_MANAGE	Maintain the asset registry, allocate and receive returns
HELPDESK_RAISE	Raise support tickets
HELPDESK_AGENT	Act on support tickets
COMPLAINT_MANAGE	Review and respond to complaints and needs
TASK_ASSIGN	Assign tasks
TASK_VIEW_ALL	View and export all employees' tasks
REPORT_VIEW	View and export reports; investigate safety incidents
DASHBOARD_EXEC	View the executive dashboard

4.4 Role to Permission Matrix

FR-RBAC-003 The system shall seed the following grants. SUPER_ADMIN shall hold every permission.

Permission	IT_EMP CV_EMP	IT_TL	IT_MGR IT_HR	CV_HR	CV_SUP	IT_FIN	IT_CEO	IT_AST CV_AST	CV_ADM
LEAVE_APPLY	•	•	•		•				
ATTENDANCE_SELF	•	•	•		•				
PAYROLL_VIEW	•		•		•				
HELPDESK_RAISE	•	•	•		•				

Permission	IT_EMP CV_EMP	IT_TL	IT_MGR IT_HR	CV_HR	CV_SUP	IT_FIN	IT_CEO	IT_AST CV_AST	CV_ADM
LEAVE_APPROVE		•	•	•	•				
ATTENDANCE_TEAM		•	•	•	•				
REPORT_VIEW		•	•	•	•	•	•		
TASK_ASSIGN		•	•						
TASK_VIEW_ALL			•						
USER_MANAGE			•	•					
EMPLOYEE_MANAGE			•	•					
TEAM_MANAGE			•	•					
CALENDAR_MANAGE			•	•					
COMPLAINT_MANAGE			•	•					
CLAIM_APPROVE			•	•					
HELPDESK_AGENT			•					•	•
ASSET_MANAGE			•	•				•	•
ORG_MANAGE			•	•					•
PAYROLL_RUN			•	•					
PAYROLL_APPROVE						•			
DASHBOARD_EXEC							•		

Note: USER_MANAGE, ORG_MANAGE and PAYROLL_RUN are seeded to the HR roles of both divisions; PAYROLL_RUN was subsequently extended to IT_MGR. TASK_VIEW_ALL, TASK_ASSIGN, HELPDESK_AGENT, CLAIM_APPROVE, COMPLAINT_MANAGE, CALENDAR_MANAGE, EMPLOYEE_MANAGE, TEAM_MANAGE and ASSET_MANAGE were added incrementally to the HR roles.

4.5 Endpoint Authorisation Rules

FR-RBAC-004 Authorisation shall be enforced on the server at the endpoint. Client-side hiding of navigation shall be treated as convenience only and shall never be the sole control.

FR-RBAC-005 The following endpoint guards shall apply:

Guard expression	Endpoints protected
USER_MANAGE	Team detachment, offboarding, leave reset, AI assistant settings and knowledge, system settings, task deletion
USER_MANAGE or EMPLOYEE_MANAGE	Employee creation, bulk creation, profile edit by id, credential set, document upload, bank add for another employee, employee deletion, password reveal
USER_MANAGE, ATTENDANCE_TEAM or DASHBOARD_EXEC	Employee directory, employee profile by id, team attendance for a date and for a range
LEAVE_APPROVE	Pending leave queue, approver queue, leave decision, bulk leave decision, permission pending and for-me queues, permission decision
LEAVE_APPROVE, USER_MANAGE or DASHBOARD_EXEC	Organisation leave calendar
ORG_MANAGE	Leave type create, update and delete; default balance allocation; master data maintenance
ORG_MANAGE or CALENDAR_MANAGE	Holiday creation and deletion
USER_MANAGE, ORG_MANAGE or TEAM_MANAGE	Team (designation) creation and deletion

Guard expression	Endpoints protected
PAYROLL_RUN	Payslip generation, salary structure upsert, payroll run creation and confirmation, payslip request inbox and decisions, logo upload, loss-of-pay preview
PAYROLL_VIEW	Salary structure read, salary list, month payslip view, another employee's payslip list
PAYROLL_APPROVE	Payroll run finance approval
PAYROLL_RUN or PAYROLL_APPROVE	Payroll run list and detail
PAYROLL_RUN, PAYROLL_VIEW, USER_MANAGE or EMPLOYEE_MANAGE	Another employee's bank accounts
ASSET_MANAGE	Asset list, create, delete, allocate, return
HELPDESK_AGENT	Agent queue, ticket status change
HELPDESK_AGENT, USER_MANAGE or DASHBOARD_EXEC	All tickets
USER_MANAGE or COMPLAINT_MANAGE	All complaints, complaint response
USER_MANAGE, DASHBOARD_EXEC or CLAIM_APPROVE	All claims, claim decision
USER_MANAGE or TASK_ASSIGN	Task creation and edit
USER_MANAGE, TASK_ASSIGN or TASK_VIEW_ALL	All tasks grouped by employee
USER_MANAGE or TASK_VIEW_ALL	Task export
REPORT_VIEW	Attendance, leave and payroll exports; safety incident list and resolution
REPORT_VIEW or USER_MANAGE	All work reports and work report export
DASHBOARD_EXEC	Executive dashboard
role SUPER_ADMIN	Community group creation and deletion, community membership management
USER_MANAGE or role IT_MGR or role IT_HR	All hourly permission requests overview

FR-RBAC-006 Where an endpoint returns records belonging to other people, the service layer shall additionally scope the result set to what the caller is entitled to see, independently of the endpoint guard.

FR-RBAC-007 A user shall never be able to decide their own request, in any workflow.

4.6 Approval Routing Hierarchy

FR-RBAC-008 Approval routing shall be single-level: for any given request, exactly one class of approver shall be able to act, so that a requester cannot choose a route that skips a level.

Leave approval routing

Applicant	Duration	Sole valid approver
Employee	up to 3 working days	a Team Leader of the applicant's own team
Employee	more than 3 working days	a Manager (IT_MGR)
Team Leader	any	a Manager (IT_MGR)
HR / Manager	any	the single designated escalation approver, identified by employee code
Administrator	any	not handled by this workflow

Hourly permission routing

Requester	Sole valid approver
Employee	a Team Leader of the requester's own team
Team Leader	HR (IT_MGR or IT_HR)
HR / Manager	the single designated escalation approver, identified by employee code

Support ticket and complaint routing

Raiser	Valid recipients
Employee or Team Leader	HR only
HR	the single designated escalation approver, identified by employee code

FR-RBAC-009 The approver picker offered to a requester shall be computed by the same rule that the decision endpoint enforces, so the list can never offer a recipient whose decision would then be rejected.

FR-RBAC-010 When a request names a specific recipient, only that recipient shall be able to act on it. When no recipient was named, the role-and-duration rule shall determine who may act.

FR-RBAC-011 Administrators shall be able to view the entire approval queue for oversight, with rows they may not decide presented as read-only.

Part II — Functional Requirements

5. Authentication and Session Management

5.1 Sign-in

FR-AUTH-001 The system shall authenticate a user by **username and password**.

FR-AUTH-002 Where the supplied identifier is not a known username, the system shall attempt a case-insensitive match against the employee's full name, and shall accept it only when exactly one employee bears that name. Ambiguous or absent matches shall be rejected.

FR-AUTH-003 Passwords shall be verified against a BCrypt hash. The stored hash shall be the sole authority for sign-in.

FR-AUTH-004 On successful sign-in the system shall reset the failed-attempt counter, clear any lock, record the last sign-in timestamp, and return an access token, a refresh token and the authenticated user's profile summary including role codes and effective permission codes.

FR-AUTH-005 A sign-in attempt against a disabled account shall be refused with an instruction to contact HR.

FR-AUTH-006 Every sign-in attempt, successful or not, shall be recorded with the identifier used, the outcome, the client IP address and the user agent. The IP shall be taken from the forwarding header's first entry when present, otherwise from the socket address. The user agent shall be truncated to a safe length.

5.2 Account Lockout

FR-AUTH-007 After **5** consecutive failed attempts the account shall be locked for **15** minutes, the counter reset, and the lock recorded on the account. Both values shall be configurable.

FR-AUTH-008 Sign-in during an active lock shall be refused with a distinct locked status, and the attempt shall still be recorded.

5.3 Tokens and Session Lifetime

FR-AUTH-009 The access token shall be a signed JWT carrying the user identifier as subject, the issuer, the username and the role codes, with issue and expiry timestamps, signed with an HMAC-SHA key derived from the configured secret.

FR-AUTH-010 The access token lifetime shall default to **4 hours** and the refresh token lifetime to **4 hours**, both configurable. The refresh window shall not silently extend a session beyond the access window.

FR-AUTH-011 A refresh token shall be an opaque, database-persisted value with an expiry and a revocation flag.

FR-AUTH-012 Exchanging a refresh token shall validate that it exists, is not revoked and is not expired; shall revoke the presented token; and shall issue a fresh token pair. Presenting a revoked or expired token shall fail with an instruction to sign in again.

FR-AUTH-013 Sign-out shall revoke **all** refresh tokens held by the user.

FR-AUTH-014 A successful password change shall revoke all refresh tokens, forcing re-authentication everywhere else.

FR-AUTH-015 The client shall attach the access token as a bearer credential to every API call, shall transparently refresh once on a 401 and retry the original request, shall coalesce concurrent refreshes into a single in-flight attempt, and shall clear stored credentials and return to sign-in when refresh fails.

5.4 Self-service Registration

FR-AUTH-016 The system shall provide a self-registration endpoint accepting username, name, password, optional date of birth, gender, national identifier, phone, e-mail, a full address block, division, and optional department, designation and office-location references.

FR-AUTH-017 Registration shall reject a duplicate username, a duplicate national identifier and a duplicate phone number, each with a distinct conflict message.

FR-AUTH-018 A self-registered account shall receive division-appropriate default self-service role — CV_EMP for the Civil division, IT_EMP otherwise — profile status ACTIVE, and a joining date of the current day.

FR-AUTH-019 Registration shall return an authenticated session, unlike administrative employee creation.

5.5 Password Management

FR-AUTH-020 A signed-in user shall be able to change their own password by supplying the current password and a new one. An incorrect current password shall be rejected.

FR-AUTH-021 Every time a password is set — at creation, at administrative reset, and at self-change — the system shall additionally store a reversibly-encrypted copy of it.

FR-AUTH-022 The reversible copy shall be encrypted with AES-GCM using a 256-bit key derived by SHA-256 over a fixed label concatenated with the application signing secret, with a fresh 12-byte random initialisation vector per value and a 128-bit authentication tag, stored Base64-encoded with the vector prefixed.

FR-AUTH-023 Failure to produce or read the reversible copy shall never prevent a password from being set or a sign-in from succeeding.

FR-AUTH-024 Accounts whose password was last set before the reversible copy existed shall report the password as not recorded until it is set again.

FR-AUTH-025 Holders of `USER_MANAGE` or `EMPLOYEE_MANAGE` shall be able to read an employee's current password through a dedicated endpoint that is called only on explicit request, so that the value never travels with an ordinary profile read, a directory listing or an export.

5.6 Availability Checks

FR-AUTH-026 The system shall expose an unauthenticated check reporting whether a username is available. **FR-AUTH-027** The system shall expose a check reporting whether a phone number is already registered.

5.7 Current User

FR-AUTH-028 The system shall expose an endpoint returning the signed-in user's identifier, employee code, username, name, national identifier, e-mail, phone, division, photo path, role codes and the distinct union of permission codes across their roles.

6. Employee Management

6.1 Administrative Employee Creation

FR-EMP-001 Holders of `USER_MANAGE` or `EMPLOYEE_MANAGE` shall be able to create an employee account with a username and password, without signing that account in, and shall receive the created account summary so the credentials can be handed over.

FR-EMP-002 Creation shall accept: username, name, password, date of birth, gender, national identifier, phone, e-mail; the full address block; division, department, designation and office-location references; reporting manager; PAN; provident-fund number; alternate phone; emergency contact and relation; blood group; personal e-mail; document paths; free-text designation, department and position titles; profile status; joining date; and an explicit role code.

FR-EMP-003 Creation shall reject a blank username, a duplicate username, a duplicate national identifier and a duplicate phone number.

FR-EMP-004 Blank optional string values shall be normalised to absent rather than stored as empty text.

FR-EMP-005 Gender shall be stored as a single upper-case character derived from the first character of the supplied value.

FR-EMP-006 Division shall be canonicalised on write: the display labels DIGITAL and INFRA shall be stored as IT and CIVIL respectively, so display labels never leak into stored data. An absent division shall default to IT.

FR-EMP-007 Profile status shall default to ACTIVE, shall be stored upper-cased, and the account shall be enabled unless the status is OFFBOARDED.

FR-EMP-008 Joining date shall default to the current day when not supplied.

FR-EMP-009 The role shall be the explicitly supplied role when it names an existing role, and otherwise the division-appropriate self-service role. An unknown role code shall be rejected.

FR-EMP-010 An employee code shall be generated automatically on creation by incrementing the highest existing code bearing the standard prefix and formatting the sequence to four digits, and shall be unique.

FR-EMP-011 Every newly created employee shall be added automatically to the organisation-wide announcement channel, so that no employee can miss a broadcast.

6.2 Bulk Employee Creation

FR-EMP-012 Holders of USER_MANAGE or EMPLOYEE_MANAGE shall be able to submit a list of employee creation requests in one call.

FR-EMP-013 Each row shall be created in its own transaction, so that one invalid row does not roll back the batch.

FR-EMP-014 The response shall report, per row, the username, the name, whether creation succeeded and the failure reason when it did not; and shall summarise how many of how many rows succeeded.

FR-EMP-015 The web application shall accept a spreadsheet workbook, shall parse sheets and columns by exact name, shall ignore unrecognised sheets and columns rather than rejecting them, shall report per sheet whether it was found and how many rows it held, and shall present the resulting credentials for distribution.

FR-EMP-016 The import screen shall display the expected workbook layout and shall offer a matching blank template for download.

6.3 Employee Directory

FR-EMP-017 Holders of USER_MANAGE, ATTENDANCE_TEAM or DASHBOARD_EXEC shall be able to browse a paged employee directory ordered by name ascending, with a default page size of 20.

FR-EMP-018 The directory shall support filtering by free-text query, division, department and profile status, with blank filter values treated as absent.

FR-EMP-019 Each directory row shall carry the identifier, employee code, name, username, e-mail, phone, division, department reference, profile status, photo path, date of birth, role codes, designation reference, designation title and technology stack.

6.4 Profile Read and Maintenance

FR-EMP-020 A signed-in user shall be able to read their own full profile.

FR-EMP-021 A signed-in user shall be able to update their own name, date of birth, gender, e-mail and full address block. Only supplied members shall be changed; absent members shall be left untouched.

FR-EMP-022 Holders of USER_MANAGE, ATTENDANCE_TEAM or DASHBOARD_EXEC shall be able to read any employee's profile by identifier.

FR-EMP-023 Holders of USER_MANAGE or EMPLOYEE_MANAGE shall be able to update any employee's profile, including identity fields, statutory identifiers, contact and emergency details, blood group, documents, free-text titles, technology stack, the full address block, division, department, designation, office location, reporting manager, employment type, joining date, profile status, employee code and role set. Only supplied members shall be changed.

FR-EMP-024 Setting profile status through an administrative update shall enable or disable the account consistently: OFFBOARDED shall disable it, any other status shall enable it.

FR-EMP-025 Replacing the role set shall resolve role codes to roles and shall replace the account's roles wholesale.

FR-EMP-026 A profile shall report: identifier, employee code, username, name, date of birth, gender, national identifier, phone, e-mail, photo path, the address block, department, designation, office-location and reporting-manager references, division, employment type, joining date, profile status, PAN, provident-fund number, alternate phone, emergency contact and relation, blood group, personal e-mail, the three free-text titles, role codes and document paths.

6.5 Credentials Administration

FR-EMP-027 Holders of `USER_MANAGE` or `EMPLOYEE_MANAGE` shall be able to set an employee's username, reset their password, or both, in one call.

FR-EMP-028 A username already held by another account shall be rejected.

FR-EMP-029 A reset password shall be at least 4 characters.

FR-EMP-030 A call supplying neither a username nor a password shall be rejected.

FR-EMP-031 A password reset shall update both the hash and the reversible copy, so that what HR can read is what the employee actually signs in with.

6.6 Photograph and Documents

FR-EMP-032 A signed-in user shall be able to upload or replace their profile photograph, and shall receive the stored path. **FR-EMP-033** A signed-in user shall be able to remove their profile photograph, after which an initials avatar shall be presented. **FR-EMP-034** Holders of `USER_MANAGE` or `EMPLOYEE_MANAGE` shall be able to upload one employee document at a time and shall receive its stored path; the caller shall collect paths and persist them on the employee as a comma-separated list.

6.7 Bank Details

FR-EMP-035 A signed-in user shall be able to list, add, update and delete their own bank accounts. **FR-EMP-036** A bank account shall carry bank name, branch name, account number, IFSC code, account holder name and a primary flag. **FR-EMP-037** The IFSC code shall be stored upper-cased. **FR-EMP-038** Marking an account primary shall demote every other account of that employee, so at most one is primary. **FR-EMP-039** Update and delete shall operate only on an account belonging to the calling user. **FR-EMP-040** Holders of `USER_MANAGE` or `EMPLOYEE_MANAGE` shall be able to add a bank account on an employee's behalf, since HR completes the joining form. **FR-EMP-041** Holders of `PAYROLL_RUN`, `PAYROLL_VIEW`, `USER_MANAGE` or `EMPLOYEE_MANAGE` shall be able to read another employee's bank accounts.

6.8 Teams

FR-EMP-042 A signed-in employee shall be able to read their own team and its members. **FR-EMP-043** The team name shall be resolved from the employee's designation title, falling back to the linked designation record's name, falling back to a generic label. **FR-EMP-044** Teammates shall be matched by designation **title** or by the designation reference, so that members carrying only a title are included. Where no teammate is found, the team shall consist of the employee alone. **FR-EMP-045** Holders of `USER_MANAGE` or role `IT_MGR` shall see the all-teams view; all other users shall see their own team only. **FR-EMP-046** Holders of `USER_MANAGE` shall be able to detach an employee from their team by clearing the designation reference.

6.9 Offboarding and Deletion

FR-EMP-047 Holders of `USER_MANAGE` shall be able to offboard an employee, supplying a relieving date, a reason and notes. **FR-EMP-048** Offboarding shall set profile status to `OFFBOARDED`, disable the account, and create an offboarding record with full-and-final status `PENDING`. **FR-EMP-049** Offboarding an already-offboarded employee shall be rejected. **FR-EMP-050** Holders of `USER_MANAGE` or `EMPLOYEE_MANAGE` shall be able to delete an employee record permanently. **FR-EMP-051** Deletion shall first clear or remove every reference to the employee that is not covered by a cascading database constraint: legacy project and team leadership references shall be nulled; the employee's own messages and

the groups they created shall be deleted; ticket assignment and comment authorship shall be nulled; and subordinates' reporting-manager reference shall be cleared. **FR-EMP-052** Each such clean-up statement shall be preceded by an existence check against the schema catalogue, so that a statement which could fail is never executed and the deletion transaction is never poisoned. **FR-EMP-053** After clean-up, pending changes shall be flushed and the employee deleted, allowing database-level cascades to remove dependent rows.

7. Organisation and Master Data

7.1 Reference Lists

FR-ORG-001 The system shall serve master data by type through a single endpoint, addressable both by path segment and by query parameter, and shall additionally accept an array of types returning a map keyed by type.

FR-ORG-002 The following types shall be supported: `blood_group`, `department`, `designation`, `employment_status`, `position`, `office_location`, `shift`, `site`. Type names shall be matched case-insensitively with hyphens treated as underscores. An unknown type shall be rejected with a clear message.

FR-ORG-003 Only active entries shall be returned, ordered by name ascending, each as an identifier and a display name.

FR-ORG-004 Designation lookups shall accept an optional division filter, canonicalising the display labels `DIGITAL` and `INFRA` to `IT` and `CIVIL`; an absent filter shall return every active designation.

FR-ORG-005 The system shall serve the list of active sites and the list of active office locations, each with its coordinates and geofence radius.

7.2 Teams as Designations

FR-ORG-006 Holders of `USER_MANAGE`, `ORG_MANAGE` or `TEAM_MANAGE` shall be able to create a team, supplying a name and an optional division. **FR-ORG-007** A blank team name shall be rejected; a name matching an existing active team, case-insensitively, shall be rejected as a conflict. **FR-ORG-008** A team code shall be derived from the name by upper-casing, replacing runs of non-alphanumeric characters with a single underscore, trimming leading and trailing underscores, and truncating to the column length. **FR-ORG-009** Team division shall default to `IT` when not supplied.

FR-ORG-010 Deleting a team by name shall first detach every member — clearing both the designation title and the designation reference — and shall then remove every designation record bearing that name, so no member is left pointing at a deleted team.

7.3 Locations, Sites and Shifts

DR-001 An office location shall carry a name, an address, a latitude, a longitude and a geofence radius in metres. **DR-002** A site shall carry a name, a code, an address, a latitude, a longitude, a geofence radius in metres and a project start date. **DR-003** A shift shall carry a name, a start time, an end time and a night-shift flag.

FR-ORG-011 The system shall be seeded with at least two office locations, three shifts covering general, morning and night patterns, and at least two project sites.

8. Attendance Management

8.1 Punch In

FR-ATT-001 A signed-in employee shall be able to punch in once per calendar day. A second punch-in on the same day shall be rejected.

FR-ATT-002 A punch-in shall record the employee, the work date, the punch-in timestamp, the mode, the supplied coordinates and the shift reference.

FR-ATT-003 The mode shall be one of `OFFICE`, `WFH`, `SITE` or `BIOMETRIC`, defaulting to `OFFICE`, and shall be stored upper-cased.

FR-ATT-004 A WFH punch shall be given status WFH and shall not be geofence-evaluated; its geofence result shall be recorded as not applicable.

FR-ATT-005 Any other mode shall be geofence-evaluated and given status PRESENT.

8.2 Geofence Evaluation

FR-ATT-006 Distance between two coordinates shall be computed by the haversine formula over a mean Earth radius of 6 371 000 metres.

FR-ATT-007 A punch shall be within the geofence when the distance from the reference centre is less than or equal to the applicable radius.

FR-ATT-008 For a SITE punch the reference shall be the supplied site, falling back to the employee's assigned site; the site reference shall be recorded on the attendance row.

FR-ATT-009 For an OFFICE or BIOMETRIC punch the reference shall be the supplied office location, falling back to the employee's assigned office location.

FR-ATT-010 Where the referenced location defines no radius, the configured default of **200 metres** shall be used.

FR-ATT-011 Where no coordinates are supplied, the punch shall be accepted, marked as outside the geofence, and flagged as a geofence exception. It shall never be blocked and shall never raise an error.

FR-ATT-012 Any punch found outside the geofence shall be flagged as a geofence exception.

8.3 Lateness

FR-ATT-013 The office start time for a punch shall be the assigned shift's start time when a shift is supplied and defines one, and otherwise the configured office start, defaulting to 09:00.

FR-ATT-014 Lateness shall be measured in whole minutes past the office start plus the configured grace period. The grace shall default to **zero minutes**, so a punch at or before the office start is on time and anything after it is late by the difference.

FR-ATT-015 The attendance row shall record both the late flag and the late minutes.

8.4 Punch Out

FR-ATT-016 A punch-out shall require an existing punch-in for the day; absence of one shall be rejected. A punch-out when one already exists shall be rejected.

FR-ATT-017 A punch-out shall record the timestamp and the supplied coordinates.

FR-ATT-018 Worked minutes shall be the whole minutes between punch-in and punch-out, floored at zero.

FR-ATT-019 Overtime shall be the whole minutes worked past the configured office end time, defaulting to 18:00. Nothing before the office end shall count. Where the punch-in itself falls after the office end, overtime shall be measured from the actual start.

8.5 Views and Summaries

FR-ATT-020 An employee shall be able to read their own attendance record for today, which shall be absent when no punch exists.

FR-ATT-021 An employee shall be able to read their own attendance rows across a date range, most recent first.

FR-ATT-022 An employee shall be able to obtain a monthly summary reporting: present days, work-from-home days, late days, absent days, total overtime minutes, total late minutes and the number of working days.

FR-ATT-023 Working days for a summary shall be counted from the first of the month up to the earlier of today and the month end, excluding Sundays and configured holidays.

FR-ATT-024 Absent days shall be the working days minus present days minus work-from-home days, floored at zero.

FR-ATT-025 Every signed-in employee shall be able to read the list of everyone absent today: enabled, non-offboarded employees who neither punched in nor hold approved or pending leave covering today. Each row shall carry the identifier, name, employee code and designation title. The list shall reflect actual punch and leave data regardless of weekday, since some teams work Saturdays.

FR-ATT-026 Every signed-in employee shall be able to read their own team's punch-in status for today, scoped to teammates sharing their designation title, each row reporting whether the teammate has punched in and when.

FR-ATT-027 Holders of ATTENDANCE_TEAM, USER_MANAGE or DASHBOARD_EXEC shall be able to read attendance for a supplied set of members on a given date, and across a date range for download.

FR-ATT-028 An attendance record shall report: identifier, employee, work date, punch-in and punch-out timestamps, mode, status, late flag, late minutes, geofence result, geofence-exception flag, worked minutes, overtime minutes, and the in and out coordinates.

8.6 Face-based Verification

FR-ATT-029 The system shall provide a face enrolment interface capturing an image from the device camera and registering it against an employee. **FR-ATT-030** The system shall provide a face verification interface capturing an image and matching it against the enrolled image for an employee. **FR-ATT-031** Face capabilities shall depend on the biometric side service; when it is unavailable the feature shall report so and shall not prevent an ordinary punch.

9. Leave Management

9.1 Leave Types and Policy

FR-LV-001 The system shall maintain leave types, each carrying a name, a unique code, a maximum days per year, a carry-forward flag, an encashable flag, an optional gender restriction, a past-dates flag, an accrual type of ANNUAL, MONTHLY or MANUAL, a minimum notice in days, a per-period limit, a paid flag and an active flag.

FR-LV-002 The system shall be seeded with Casual Leave, Sick Leave, Earned Leave, Compensatory Off, Maternity Leave, Paternity Leave and Loss of Pay, with entitlements, restrictions and notice periods appropriate to each.

FR-LV-003 Any signed-in user shall be able to list active leave types, ordered by name.

FR-LV-004 Holders of ORG_MANAGE shall be able to create, update and delete leave types.

FR-LV-005 Leave type codes shall be unique, matched case-insensitively.

FR-LV-006 Creating a type whose code matches a **soft-deleted** type shall reactivate and update that type rather than fail on the uniqueness constraint. Creating one whose code matches an **active** type shall be rejected.

FR-LV-007 Updating a type shall reject a code already held by a different type.

FR-LV-008 Deleting a leave type shall be a soft delete, marking it inactive and preserving history.

9.2 Balances and Allocation

FR-LV-009 A leave balance shall be held per employee, per leave type, per year, carrying an allocated quantity and a used quantity; the available quantity shall be the difference.

FR-LV-010 An employee shall be able to read their own balances for a year, defaulting to the current year, each row reporting the type name and code alongside the allocated, used and available quantities.

FR-LV-011 Holders of ORG_MANAGE shall be able to allocate default annual balances to every enabled employee in one action, using each active type's configured maximum days per year as the allocated quantity.

FR-LV-012 Types with no cap or a zero cap shall be skipped by bulk allocation.

FR-LV-013 Bulk allocation shall never overwrite an existing balance row, so that it is safe to run repeatedly, and shall report how many rows were created, how many employees were covered and for which year.

FR-LV-014 Allocation shall be permitted only for the current year or a later one; a year already past shall be rejected with a clear message.

FR-LV-015 Holders of `USER_MANAGE` shall be able to reset an employee's leave, zeroing every used quantity and clearing their leave request history.

9.3 Working-day Calculation

FR-LV-016 Working days within a date range shall be counted excluding **Sundays** and configured **holidays**. Saturday shall count as a working day.

9.4 Applying for Leave

FR-LV-017 An employee shall be able to apply for leave supplying a leave type, a from date, a to date, a reason, an optional attachment path and an optional named approver.

FR-LV-018 An end date earlier than the start date shall be rejected.

FR-LV-019 Where the leave type carries a gender restriction and the applicant's gender is known, a mismatch shall be rejected with a message stating the type is not applicable to their profile.

FR-LV-020 Unless the type explicitly allows past dates, a start date before today shall be rejected.

FR-LV-021 Where the type defines a minimum notice and the start date is in the future, an application with less notice than required shall be rejected, naming the required notice.

FR-LV-022 The working days of the request shall be computed and stored. A range containing no working days shall be rejected.

FR-LV-023 Where the type defines a per-period limit, the limit shall be applied **per calendar quarter**: the number of existing requests for that type whose start date falls within the quarter containing the requested start date shall not exceed the limit. Exceeding it shall be rejected with a message naming the allowance and the date from which the next one becomes available.

FR-LV-024 For every type other than Loss of Pay, a balance row for the applicant, the type and the year of the start date shall be required; its absence shall be rejected. The available quantity shall be at least the requested working days; a shortfall shall be rejected, naming both figures.

FR-LV-025 A submitted request shall be created with status `PENDING` and shall record the applicant, the type, the dates, the working days, the reason, the attachment path and the named approver.

9.5 Approval Notification

FR-LV-026 On submission, where a specific approver was named, only that person shall be notified. Where none was named, every holder of `LEAVE_APPROVE` shall be notified.

FR-LV-027 Each notification shall name the applicant, the leave type, the number of working days and the date or date range, shall be recorded in the notification feed, shall be pushed live, and shall link to the approvals screen.

FR-LV-028 Each notified approver holding a usable mobile number shall additionally receive an SMS.

9.6 Own Requests and Cancellation

FR-LV-029 An employee shall be able to browse their own leave requests, paged, most recent first, each row naming the applicant and the leave type.

FR-LV-030 An employee shall be able to cancel their own request. Cancelling another employee's request shall be rejected.

FR-LV-031 Only a `PENDING` or `APPROVED` request shall be cancellable.

FR-LV-032 Cancelling an approved request shall refund the consumed balance, floored at zero, for every type other than Loss of Pay.

FR-LV-033 Cancellation shall notify whoever was handling the request — the named approver, or failing that the decider — unless that person is the applicant, stating who cancelled, whether it had been approved, the type and the dates.

9.7 Approver Queues

FR-LV-034 Holders of `LEAVE_APPROVE` shall be able to read the pending queue. Rows shall be included where the approver may act; administrators shall additionally see the entire queue, with rows they may not decide marked read-only.

FR-LV-035 Holders of `LEAVE_APPROVE` shall be able to read their full queue across all statuses, supporting pending, approved and rejected views, most recent first, resolving in one pass the names of applicants, named approvers and deciders.

FR-LV-036 A queue row shall additionally report the applicant's highest leave-routing role, whether the viewer may act on it, the named approver's name, the decider's name and the applicant's designation title.

FR-LV-037 An employee shall be able to obtain the list of approvers valid for a leave of a given number of days, computed by the same routing rule the decision endpoint enforces, each entry carrying the identifier, name and employee code.

FR-LV-038 Every signed-in user shall be able to read who is on approved leave today — where today falls within the from and to dates — each row naming the person, their designation title and their employee code.

9.8 Organisation Leave Calendar

FR-LV-039 Holders of `LEAVE_APPROVE`, `USER_MANAGE` or `DASHBOARD_EXEC` shall be able to read all leave overlapping a date range.

FR-LV-040 Calendar visibility shall be scoped: holders of `USER_MANAGE`, administrators, managers and HR shall see the whole organisation; a Team Leader shall see only their own team and themselves.

9.9 Decisions

FR-LV-041 Holders of `LEAVE_APPROVE` shall be able to decide a single request, and shall be able to decide many at once.

FR-LV-042 A bulk decision shall act only on requests still `PENDING` and shall report how many were decided.

FR-LV-043 A decision on a request that is not `PENDING` shall be rejected, naming its current state.

FR-LV-044 A decision shall be rejected where the decider is not the person entitled to act on that request under the routing rule, so that the API enforces routing independently of the screen.

FR-LV-045 A decision shall be either `APPROVED` or `REJECTED`; any other value shall be rejected.

FR-LV-046 A rejection shall require a reason; a blank reason shall be rejected.

FR-LV-047 On approval, for every type other than Loss of Pay, the balance row shall be required and the available quantity re-checked at decision time; an intervening shortfall shall be rejected. The used quantity shall then be increased by the request's working days.

FR-LV-048 A decision shall record the outcome, the decider, the decision timestamp and the comment.

FR-LV-049 The applicant shall be notified of the outcome in the feed, pushed live, linking to the leave screen; and shall additionally receive an SMS naming the type, the dates, the outcome and any comment.

9.10 Loss-of-Pay Preview

FR-LV-050 Holders of `PAYROLL_RUN` shall be able to obtain a loss-of-pay preview for an employee and a month, reporting: unpaid approved leave days, paid approved leave days, total leave days, the number of approved leave

requests, days actually attended, days neither worked nor covered by leave, and the number of working days in the month.

FR-LV-051 Leave days shall be classified paid or unpaid by the leave type's paid flag.

FR-LV-052 Attended days shall count both PRESENT and WFH records.

10. Hourly Permission Requests

FR-PRM-001 An employee shall be able to request a short absence during a working day, supplying a date, a start time, an end time in HH:mm, a reason and an optional named approver.

FR-PRM-002 Unparseable times shall be rejected with an instruction to use HH:mm. An end time not after the start time shall be rejected.

FR-PRM-003 The duration in hours shall be computed from the minute difference, divided by sixty and rounded half-up to two decimal places.

FR-PRM-004 A submitted request shall be created with status PENDING.

FR-PRM-005 Where an approver was named, that person shall be notified in the feed, pushed live, linking to the permissions screen, and shall receive an SMS where a usable number exists; the message shall name the requester, the hours, the date and the time window.

FR-PRM-006 A requester shall be able to list their own requests, most recent first.

FR-PRM-007 A requester shall be able to obtain the list of approvers valid for them, computed by the routing rule of §4.6.

FR-PRM-008 Holders of LEAVE_APPROVE shall be able to read the pending requests addressed to them, and the full history of requests addressed to them across all statuses.

FR-PRM-009 Holders of USER_MANAGE, or role IT_MGR, or role IT_HR, shall be able to read every request as a read-only overview, most recent first.

FR-PRM-010 Holders of LEAVE_APPROVE shall be able to approve or reject a request. A rejection shall require a reason.

FR-PRM-011 A pending request whose date has already passed shall no longer be decidable; the attempt shall be rejected with a message stating the date and that the request is now overdue.

FR-PRM-012 A decision shall record the outcome, the decider, the timestamp and the comment; and shall notify the requester in the feed, pushed live, and by SMS where a usable number exists.

FR-PRM-013 A requester shall be able to cancel their own request while it is still PENDING; cancelling another's request or a decided request shall be rejected. Cancellation shall remove the request.

FR-PRM-014 A request row shall report the requester's name, employee code and designation title, the named approver's name and the decider's name.

11. Payroll and Payslips

11.1 Salary Structures

FR-PAY-001 Holders of PAYROLL_RUN shall be able to create or replace an employee's active salary structure, supplying basic salary, house-rent allowance, other allowances, a provident-fund amount, an ESI-applicability flag and a professional-tax amount.

FR-PAY-002 Absent monetary members shall default to zero; ESI applicability shall default to applicable.

FR-PAY-003 An upsert shall reuse the employee's existing active structure where one exists, shall mark the structure active, and shall set an effective-from date of today when none is present.

FR-PAY-004 Holders of `PAYROLL_VIEW` shall be able to read one employee's active structure and to list every active structure, each reporting the employee's name and code, the components and the computed gross.

FR-PAY-005 The provident-fund figure on a structure shall be treated as a **flat rupee amount**, not a percentage.

11.2 Direct Payslip Generation

FR-PAY-006 Holders of `PAYROLL_RUN` shall be able to generate a payslip for an employee, a month and a year, supplying overtime hours, performance pay, tax deducted at source, an advance deduction, a manual loss-of-pay deduction and other deductions.

FR-PAY-007 Generation shall be rejected for a month earlier than the employee's joining month, naming the joining date and the earliest permissible month.

FR-PAY-008 Generation shall require an active salary structure; its absence shall be rejected, naming the employee.

FR-PAY-009 Generation shall reuse an existing payslip row for the same employee, month and year where one exists.

FR-PAY-010 Earnings shall be computed as follows:

- overtime pay = (basic + HRA + allowances) ÷ 240, rounded half-up, multiplied by the overtime hours;
- gross = basic + HRA + allowances + overtime pay + performance pay, floored at zero, scaled to two places.

FR-PAY-011 Deductions shall be computed as follows:

- provident fund = the structure's flat amount;
- ESI = 0.75 % of gross, but only where the structure marks ESI applicable **and** gross does not exceed the wage ceiling of 21 000; otherwise zero;
- professional tax = the structure's amount;
- tax deducted at source, advance and other deductions as supplied;
- the manual loss-of-pay deduction shall be grouped with other deductions for storage;
- total deductions = provident fund + ESI + professional tax + tax + other + advance.

FR-PAY-012 Net pay shall be gross minus total deductions, scaled to two places.

FR-PAY-013 Attendance-driven automatic loss of pay shall not be deducted; loss of pay shall be entered manually, so that the two cannot double-count.

FR-PAY-014 A generated payslip shall be auto-completed from the employee's record: the primary bank account's name and number, or the first account, or a placeholder where none exists; the designation title, falling back to the linked designation's name, falling back to a placeholder; the department title with the same fallback chain; the pay date set to today; and the working days set to the number of days in the month.

FR-PAY-015 The payslip shall additionally record a computed loss-of-pay **day count**: working days in the month, up to today where the month is current, excluding Sundays and holidays, on which the employee has no punch-in.

FR-PAY-016 Generation shall render a PDF and persist its path on the payslip.

11.3 Payslip Request Workflow

FR-PAY-017 Any signed-in user shall be able to raise a payslip request for a month and a year with an optional note.

FR-PAY-018 A request for a month earlier than the requester's joining month shall be rejected, naming the joining date and the earliest permissible month.

FR-PAY-019 A second pending request for the same month shall be rejected, naming the month.

FR-PAY-020 A raised request shall be created with status `PENDING` and shall notify every holder of `PAYROLL_RUN` in the feed, pushed live, linking to the payroll requests screen, and by SMS.

FR-PAY-021 A requester shall be able to list their own requests, most recent first.

FR-PAY-022 Holders of `PAYROLL_RUN` shall be able to read the request inbox, either pending only or complete, most recent first, each row naming the requester and their employee code.

FR-PAY-023 Holders of PAYROLL_RUN shall be able to reject a request with a note. A request not PENDING shall be rejected, naming its state. Rejection shall record the outcome, the decider, the timestamp and the note, and shall notify the requester in the feed and by SMS, quoting the note.

FR-PAY-024 Holders of PAYROLL_RUN shall be able to approve a request by completing a payslip form. Approval shall be rejected where the request is not PENDING.

FR-PAY-025 Approval shall accept, as fully admin-entered values: basic salary, house-rent allowance, allowances, overtime pay, performance pay, expenses pay; provident fund, ESI, professional tax, tax deducted at source, health insurance, salary advance and other deductions; the loss-of-pay day count; and optional overrides for company name, logo, tax registration number and address, bank name and account, designation, department, working days, pay date, employee name, employee code and a decision note.

FR-PAY-026 On approval, gross shall be the sum of the six earning members; total deductions the sum of the seven deduction members, with salary advance treated as a deduction line; and net pay the difference. All shall be scaled to two places.

FR-PAY-027 Approval shall reuse an existing payslip row for the month where one exists, shall mark the payslip's source as request-driven, shall store every admin-entered value so the download reproduces exactly that format, shall render the PDF using the entered display name and code, shall link the payslip to the request, and shall close the request as APPROVED recording the decider, the timestamp and the note.

FR-PAY-028 Approval shall notify the requester that the payslip is ready to download, in the feed and by SMS.

FR-PAY-029 Holders of PAYROLL_RUN shall be able to upload a company logo for use on payslips.

11.4 Payslip Access

FR-PAY-030 An employee shall be able to list their own payslips, most recent year and month first. **FR-PAY-031** Holders of PAYROLL_VIEW shall be able to list another employee's payslips and to read the payslip summary map for a month keyed by employee. **FR-PAY-032** A payslip shall be readable by its owner; a non-privileged caller requesting another employee's payslip shall be rejected. **FR-PAY-033** A payslip PDF shall be downloadable by its owner, and by privileged callers. A non-privileged caller requesting another employee's PDF shall be rejected. **FR-PAY-034** A payslip PDF shall be regenerated at download time, so that corrected designation, department and bank metadata are always reflected.

11.5 Payslip Document

FR-PAY-035 The payslip PDF shall carry the company logo, name, address and tax registration number; a metadata grid identifying the employee, code, designation, department, pay period, pay date, working days, loss-of-pay days, bank name and account; an earnings and deductions table with totals; the net pay rendered in words; and a signature block naming the employer signatory, their title and contact.

FR-PAY-036 The PDF shall embed fonts capable of rendering the currency symbol and Indian-language text.

11.6 Payroll Runs

FR-PAY-037 Holders of PAYROLL_RUN shall be able to create a payroll run for a month and a year. A second run for the same month shall be rejected.

FR-PAY-038 A created run shall be recorded with status PREVIEW, the initiator and the run timestamp.

FR-PAY-039 A run shall attempt to generate a payslip for every enabled employee and shall link each generated payslip to the run. Employees for whom generation is not possible — for example those without an active salary structure — shall be skipped without failing the run.

FR-PAY-040 Holders of PAYROLL_RUN shall be able to confirm a run. A run not in PREVIEW shall be rejected. Confirmation shall move it to CONFIRMED.

FR-PAY-041 Holders of PAYROLL_APPROVE shall be able to give finance approval to a run. A run not in CONFIRMED shall be rejected. Approval shall move it to FINANCE_APPROVED and shall record the approver and the timestamp.

FR-PAY-042 Finance approval shall notify every employee whose payslip belongs to the run that their payslip for that month is ready, linking to the payslips screen.

FR-PAY-043 Holders of `PAYROLL_RUN` or `PAYROLL_APPROVE` shall be able to list runs, most recent period first, and to read one run with its payslips, each naming the employee and their code.

FR-PAY-044 The run status sequence shall be `PREVIEW` → `CONFIRMED` → `FINANCE_APPROVED` → `PAID`.

12. Task Management

12.1 Assignment

FR-TSK-001 Holders of `USER_MANAGE` or `TASK_ASSIGN` shall be able to assign a task, supplying a title, a description, an assignee, a due date, a priority and optional team-batch identity.

FR-TSK-002 Assignment scope shall be enforced by capability:

- a holder of `USER_MANAGE` shall be unrestricted;
- a holder of `TASK_VIEW_ALL` without `USER_MANAGE` — that is, HR — shall be able to assign **only to Team Leaders**, of any team, and an attempt otherwise shall be rejected with that message;
- any other holder of `TASK_ASSIGN` — that is, a Team Leader — shall be able to assign **only within their own team**, matched by designation title, and an attempt otherwise shall be rejected with that message.

FR-TSK-003 A due date earlier than today shall be rejected.

FR-TSK-004 Priority shall be one of `LOW`, `MEDIUM` or `HIGH`, matched case-insensitively, defaulting to `MEDIUM` for any other value.

FR-TSK-005 A task shall be created with status `PENDING`, recording the assignee, the assigner and the title trimmed of surrounding whitespace.

FR-TSK-006 Where a team batch identifier is supplied, it shall be recorded together with an optional team name, so the members of one team assignment remain identifiable as a group.

FR-TSK-007 Assignment shall notify the assignee in the feed, pushed live, linking to the tasks screen; and shall send an SMS naming the assignee, the task title, their employee code and the due date where one exists.

12.2 Progress and Completion

FR-TSK-008 An assignee shall be able to mark their own task complete. Completing another employee's task shall be rejected.

FR-TSK-009 Completion shall set status `COMPLETED`, progress to 100 and the completion timestamp, and shall be idempotent.

FR-TSK-010 Completion shall notify the assigner in the feed and by SMS, naming who completed which task.

FR-TSK-011 An assignee shall be able to report progress between 0 and 100 on their own task; values outside the range shall be clamped. Reporting progress on another employee's task shall be rejected.

FR-TSK-012 Progress shall drive status: 100 shall mean `COMPLETED` with a completion timestamp; any value above zero shall mean `IN_PROGRESS`; zero shall mean `PENDING`. Moving away from completion shall clear the completion timestamp.

FR-TSK-013 Task state shall be **forward-only**. A move to an earlier state — completed back to in-progress or pending, or in-progress back to pending — shall be rejected with a message naming both states. The rule shall be enforced by the API and not only by the interface.

FR-TSK-014 Progress reporting shall notify the assigner of the new percentage in the feed.

12.3 Editing and Removal

FR-TSK-015 Holders of `USER_MANAGE` or `TASK_ASSIGN` shall be able to edit a task's title, description, priority, due date and status. A non-administrator shall be able to edit only tasks they assigned; an attempt otherwise shall be rejected.

FR-TSK-016 An edit that changes status shall obey the forward-only rule. Where a task is set to in-progress and its progress is zero or already complete, progress shall be set to a mid-point value; otherwise the assignee's reported progress shall be preserved.

FR-TSK-017 An edit shall notify the assignee in the feed and by SMS, naming the editor, the task and the due date where one exists.

FR-TSK-018 Holders of `USER_MANAGE` shall be able to delete a task, and to delete every member task belonging to one team assignment by its batch identifier. Deleting a batch that holds no tasks shall report not found.

12.4 Views and Export

FR-TSK-019 An employee shall be able to list their own tasks, most recent first, each naming the assigner.

FR-TSK-020 Holders of `USER_MANAGE`, `TASK_ASSIGN` or `TASK_VIEW_ALL` shall be able to read tasks grouped by employee, most recent first within each group, each group reporting the employee's name, code and division together with total, pending and completed counts.

FR-TSK-021 The grouped view shall be scoped: holders of `USER_MANAGE` or `TASK_VIEW_ALL` shall see everyone; any other viewer — that is, a Team Leader — shall see only employees sharing their designation title.

FR-TSK-022 The grouped view shall support filtering by division, canonicalising display labels, and by a free-text query matched against the employee's name or code.

FR-TSK-023 Holders of `USER_MANAGE` or `TASK_VIEW_ALL` shall be able to export tasks to a spreadsheet, filtered by division and by an assigned-date window, with columns: employee, employee code, team, task, description, status, priority, due date, assigned date and assigned by.

FR-TSK-024 The exported team name shall be the batch team name where present, otherwise the employee's designation title.

FR-TSK-025 Export column widths shall be auto-sized.

13. Work Reports

FR-WRP-001 An employee shall be able to record a daily work report carrying a work date, a project name, hours worked and a task or module description.

FR-WRP-002 Absent hours shall be stored as zero.

FR-WRP-003 An employee shall be able to list their own reports, most recent date first and, within a date, most recent entry first.

FR-WRP-004 An employee shall be able to edit and delete their own reports; editing or deleting another employee's report shall be rejected.

FR-WRP-005 Holders of `REPORT_VIEW` or `USER_MANAGE` shall be able to read all reports grouped by employee, most recent first within each group, each group reporting the employee's name, code, the number of rows and the sum of hours.

FR-WRP-006 The grouped view shall support a free-text query matched against the employee's name or code.

FR-WRP-007 A Team Leader shall be able to read the grouped view restricted to their own team, matched by designation title; another team's reports shall never be exposed.

FR-WRP-008 Holders of `REPORT_VIEW` or `USER_MANAGE` shall be able to export all reports to a spreadsheet, optionally filtered by a date window, with columns: date, employee, employee code, team, project, hours and task or module.

14. Travel and Expense Claims

14.1 Travel Claims

FR-CLM-001 An employee shall be able to submit a travel claim carrying a date, a location, a category, starting and ending odometer readings, total kilometres, hills and plains kilometres, a computed total amount, bus fare, other amounts, a gross total, remarks, a fuel-receipt path and additional photograph paths.

FR-CLM-002 A submitted claim shall be created with status PENDING.

FR-CLM-003 Submission shall notify every approver — the union of CLAIM_APPROVE and USER_MANAGE holders, de-duplicated — in the feed, pushed live, linking to the claims screen, and by SMS, naming the submitter, the claim date and the gross total.

FR-CLM-004 An employee shall be able to upload claim attachments and shall receive their stored paths.

FR-CLM-005 An employee shall be able to correct their own claim **while it is still pending**; correcting a reviewed claim shall be rejected with that explanation, and correcting another employee's claim shall be rejected as access denied.

FR-CLM-006 Holders of CLAIM_APPROVE, USER_MANAGE or DASHBOARD_EXEC shall be able to correct any claim regardless of its decision, since a wrong amount otherwise stays wrong and they answer for it.

FR-CLM-007 Where the owner corrects a claim, every approver shall be notified that it changed. Where somebody else corrects it, the owner shall be notified in the feed and by SMS, naming who corrected it, the date and the amount.

FR-CLM-008 An employee shall be able to list their own claims, most recent date first.

FR-CLM-009 A Team Leader shall be able to list claims raised by their own team, matched by designation title, most recent date first; other teams' claims shall never be exposed. Where the caller has no team, only their own claims shall be returned.

FR-CLM-010 Holders of USER_MANAGE, DASHBOARD_EXEC or CLAIM_APPROVE shall be able to list every claim.

FR-CLM-011 Holders of USER_MANAGE, DASHBOARD_EXEC or CLAIM_APPROVE shall be able to set a claim's status to APPROVED, REJECTED or PENDING; any other value shall be rejected. A rejection shall require a reason.

FR-CLM-012 A decision shall record the status, the comment, the decider and the decision timestamp; and, where the status is not pending, shall notify the owner of the outcome and the reason in the feed and by SMS.

FR-CLM-013 An employee shall be able to delete their own claim; an administrator shall be able to delete any claim; any other deletion shall be rejected as access denied.

FR-CLM-014 A claim row shall report the owner's name, employee code and designation title, the decision comment, the decider's name and the decision timestamp.

FR-CLM-015 The system shall hold configurable per-kilometre rates for hills and plains travel as system settings.

FR-CLM-016 The system shall be able to render a printable claim invoice.

14.2 General Expense Claims

FR-CLM-017 An employee shall be able to submit a general expense claim carrying a category, an amount, a claim date and a receipt path, created with manager and finance statuses both PENDING. **FR-CLM-018** An employee shall be able to list their own general expense claims, most recent first. **FR-CLM-019** The system shall be able to list general expense claims awaiting either a manager or a finance decision. **FR-CLM-020** The system shall support a manager decision and a finance decision on a general expense claim, recorded independently.

15. Asset Management

FR-AST-001 Holders of `ASSET_MANAGE` shall be able to register an asset carrying a category of `IT`, `INFRA` or `MACHINERY`, an asset type, a brand, a model, a serial number, a registration number, a purchase date and cost, warranty and maintenance-contract expiry dates, a site reference, a depreciation rate and a quantity.

FR-AST-002 An asset code shall be generated automatically with a category-derived prefix — `INF` for infrastructure, `MCH` for machinery, `AST` otherwise — followed by the current year and a five-digit sequence, and shall be unique.

FR-AST-003 Quantity shall default to one. An asset registered with a quantity of zero or less shall be `OUT_OF_STOCK`; otherwise `IN_STOCK`.

FR-AST-004 A QR tag shall be generated and attached on registration, and shall be downloadable as a PNG image encoding the asset code with an identifying prefix, at a size suitable for label printing.

FR-AST-005 Holders of `ASSET_MANAGE` shall be able to browse assets, paged, filtered by status or by category.

FR-AST-006 Any signed-in user shall be able to read one asset by identifier and to look an asset up by its code.

FR-AST-007 Holders of `ASSET_MANAGE` shall be able to allocate an asset to an employee. Allocation shall be rejected where the asset is out of stock, where its quantity is exhausted, or where it is retired or lost, naming the reason.

FR-AST-008 Allocation shall create an allocation record with an allocation timestamp, shall decrement the quantity, shall mark the asset out of stock when the quantity reaches zero, shall record the assignee and shall update the modification timestamp.

FR-AST-009 Allocation shall notify the assignee in the feed and by SMS, naming the asset type and code and requesting acknowledgement of receipt.

FR-AST-010 The assignee shall be able to acknowledge receipt. Acknowledgement shall require an active allocation and shall be rejected for anybody other than the assignee. It shall record the acknowledgement and its timestamp.

FR-AST-011 Holders of `ASSET_MANAGE` shall be able to record a return with a condition. A return shall require an active allocation.

FR-AST-012 A return shall set the return timestamp and condition, shall increment the quantity, shall clear the assignee, and shall set the asset status from the condition: `LOST` for a lost item, `UNDER_REPAIR` for a damaged item, `IN_STOCK` otherwise. An absent condition shall be treated as good.

FR-AST-013 An employee shall be able to list the assets currently allocated to them.

FR-AST-014 Holders of `ASSET_MANAGE` shall be able to delete an asset; allocation history shall be removed first so that referential constraints do not block the deletion.

FR-AST-015 The system shall hold a maintenance log per asset carrying a fault description, a vendor, an estimated turnaround, a cost, a status, a scheduled date and preventive intervals in days and hours.

FR-AST-016 The system shall hold a software licence register carrying a product name, a vendor, a licence key, a licence type, seats purchased and consumed, and an expiry date.

FR-AST-017 Asset statuses shall be `IN_STOCK`, `ASSIGNED`, `OUT_OF_STOCK`, `UNDER_REPAIR`, `RETIRED`, `LOST`, `DEPLOYED` and `BREAKDOWN`.

16. Support Helpdesk

FR-HD-001 Any signed-in user shall be able to raise a support ticket carrying a title, a description, attachments, a type, a priority and an addressed recipient.

FR-HD-002 A ticket code shall be generated as a fixed prefix, the current year and a five-digit sequence, and shall be unique.

FR-HD-003 The ticket type shall default to `IT` and shall be stored upper-cased. The priority shall default to `MEDIUM` and shall be stored upper-cased.

FR-HD-004 The ticket **category shall be derived from the raiser's division**, not chosen manually: **Infra** for Civil staff, **Digital** for IT staff.

FR-HD-005 A ticket shall be created with status **OPEN** and an SLA due time derived from the priority: 4 hours for critical, 8 hours for high, 24 hours for medium and 48 hours otherwise.

FR-HD-006 Where the ticket names a recipient other than the raiser, that person shall be notified in the feed, pushed live, linking to the helpdesk screen, and by SMS, naming the raiser, the code and the title.

FR-HD-007 Any signed-in user shall be able to obtain the list of valid recipients for their own ticket, computed by the routing rule of §4.6, each entry carrying the identifier, name, employee code and designation. Nobody shall appear in their own list.

FR-HD-008 A raiser shall be able to correct their own ticket's title, description, attachments, type, priority and recipient **only while it is still open**. Once an agent has begun work the attempt shall be rejected with a message naming the current state and directing the raiser to add a reply instead.

FR-HD-009 The ticket code, the category and the status shall not be alterable by the raiser.

FR-HD-010 Changing the priority shall recompute the SLA due time.

FR-HD-011 Correcting a ticket shall notify its current recipient that the details changed; where the recipient was changed, the previous recipient shall also be notified that it was handed on.

FR-HD-012 A raiser shall be able to browse their own tickets, paged, most recent first.

FR-HD-013 Holders of **HELPDESK_AGENT** shall be able to browse their queue — tickets addressed to them, or every ticket of a given status when a status filter is supplied — paged, most recent first.

FR-HD-014 Holders of **HELPDESK_AGENT**, **USER_MANAGE** or **DASHBOARD_EXEC** shall be able to browse all tickets, paged, optionally filtered by status.

FR-HD-015 Any signed-in user shall be able to read one ticket with its comment thread in chronological order.

FR-HD-016 Any signed-in user shall be able to add a comment with an optional attachment. The counterparty — the recipient when the author is the raiser, the raiser otherwise — shall be notified in the feed and by SMS, unless they are the author.

FR-HD-017 Holders of **HELPDESK_AGENT** shall be able to change a ticket's status. The raiser shall never be able to decide their own ticket; the attempt shall be rejected with an explanation that the person it was sent to will handle it.

FR-HD-018 Where a ticket names a recipient, only that recipient shall be able to move it along; another agent's attempt shall be rejected. A ticket predating recipient recording shall remain open to any agent.

FR-HD-019 Valid statuses shall be **OPEN**, **IN_PROGRESS**, **AWAITING_PARTS**, **RESOLVED** and **CLOSED**; any other value shall be rejected.

FR-HD-020 Status transitions shall move forward only, one step at a time, with the single exception that **IN_PROGRESS** may move either to **AWAITING_PARTS** or directly to **RESOLVED**. Any other transition shall be rejected, naming both states.

FR-HD-021 A status change may reassign the ticket. Reaching **RESOLVED** or **CLOSED** shall set the resolution timestamp.

FR-HD-022 A status change shall notify the raiser in the feed and by SMS, naming the ticket code and the new state.

FR-HD-023 Only the raiser shall be able to rate a ticket, and only once it is resolved or closed; other attempts shall be rejected.

FR-HD-024 A ticket row shall report the raiser's name and employee code, the recipient's name and, on the detail view, the comment thread with each author's name.

FR-HD-025 Any signed-in user shall be able to upload ticket attachments and shall receive their stored paths.

17. Complaints and Needs

FR-CMP-001 Any signed-in user shall be able to submit either a **complaint** or a **need**, carrying a category, a subject, a description, a priority and an addressed recipient.

FR-CMP-002 The kind shall be `COMPLAINT` or `NEED`, defaulting to complaint. The priority shall be `LOW`, `MEDIUM` or `HIGH`, defaulting to medium. Unrecognised values shall fall back to the default rather than being rejected.

FR-CMP-003 Subject and description shall be trimmed of surrounding whitespace; a blank category shall be stored as absent.

FR-CMP-004 A reference code shall be generated as a fixed prefix, the current year and a five-digit sequence, derived by **incrementing the highest existing code for the year** rather than by counting rows, so that deleting a row can never regenerate a used code.

FR-CMP-005 A submission shall be created with status `OPEN`.

FR-CMP-006 Where a recipient was named, only that person shall be notified. Where none was named, every HR and administrator — the de-duplicated union of `COMPLAINT_MANAGE` and `USER_MANAGE` holders — shall be notified. The submitter shall never be notified of their own submission.

FR-CMP-007 Notification shall be recorded in the feed, pushed live, linked to the complaints screen, and sent by SMS, naming the submitter, the kind and the reference code.

FR-CMP-008 Any signed-in user shall be able to obtain the list of valid recipients, computed by the routing rule of §4.6, each entry carrying the identifier, name, employee code and a role label; somebody holding both HR and administrative capability shall be labelled with the wider of the two. The requester shall be excluded. This list shall be available to every signed-in user because the full directory is not theirs to read.

FR-CMP-009 A submitter shall be able to browse their own submissions, paged, most recent first.

FR-CMP-010 Holders of `USER_MANAGE` or `COMPLAINT_MANAGE` shall be able to browse all submissions, paged, optionally filtered by status and by kind, and to read one submission.

FR-CMP-011 Holders of `USER_MANAGE` or `COMPLAINT_MANAGE` shall be able to respond, setting a status of `OPEN`, `IN_REVIEW`, `RESOLVED` or `REJECTED` — any other value shall be rejected — and recording a response, the handler and the update timestamp.

FR-CMP-012 Reaching `RESOLVED` or `REJECTED` shall set the resolution timestamp; returning to an earlier status shall clear it.

FR-CMP-013 A response shall notify the submitter of the new state in the feed and by SMS.

FR-CMP-014 A submission row shall report the submitter's name and employee code, their team, the handler's name and the named recipient's name.

18. Safety Incident Management

FR-SAF-001 Any signed-in user shall be able to report a safety incident carrying an incident type, a description, a zone, an anonymity flag, a severity and an occurrence timestamp.

FR-SAF-002 The incident type shall be one of `NEAR_MISS`, `MINOR_INJURY`, `MAJOR_INJURY`, `PROPERTY_DAMAGE` or `ENV_HAZARD`, defaulting to near miss. The severity shall be `LOW`, `MEDIUM`, `HIGH` or `CRITICAL`, defaulting to medium. Unrecognised values shall fall back to the default.

FR-SAF-003 A reference code shall be generated as a fixed prefix, the current year and a five-digit sequence.

FR-SAF-004 A report shall be created with status `OPEN`; the description shall be trimmed and a blank zone stored as absent.

FR-SAF-005 Every holder of `REPORT_VIEW` other than the reporter shall be notified in the feed, pushed live, linking to the safety screen, and by SMS, naming the reference code and the incident type in readable form.

FR-SAF-006 Where the report is marked anonymous, the reporter shall be presented as "Anonymous" in every notification and in every view.

FR-SAF-007 A reporter shall be able to browse their own reports, paged, most recent first.

FR-SAF-008 Holders of `REPORT_VIEW` shall be able to browse all incidents, paged, optionally filtered by status and by incident type, and to read one incident.

FR-SAF-009 Holders of `REPORT_VIEW` shall be able to record an investigation outcome, setting a status of `OPEN`, `INVESTIGATING`, `RESOLVED` or `CLOSED` — any other value shall be rejected — with resolution notes, recording the resolver.

FR-SAF-010 Reaching `RESOLVED` or `CLOSED` shall set the resolution timestamp; returning to an earlier status shall clear it.

FR-SAF-011 An update shall notify the reporter of the new state in the feed and by SMS, unless the report was anonymous and no reporter is recorded.

19. Onboarding and Offboarding

FR-ONB-001 Holders of `USER_MANAGE` shall be able to start onboarding for an employee. Starting it twice shall be rejected.

FR-ONB-002 Starting onboarding shall create a checklist with status `IN_PROGRESS` and a start timestamp, and shall populate it with the standard tasks: Document Collection, Bank Details, IT Asset Assignment, Team Introduction and Security Training.

FR-ONB-003 The system shall be able to list the employees currently in onboarding — those holding a checklist in progress.

FR-ONB-004 The system shall be able to read an employee's checklist with its tasks, each reporting its name, description, completion state and completion timestamp.

FR-ONB-005 A checklist task shall be markable complete, recording the completion timestamp. Completing a task that does not belong to the employee's checklist shall be rejected.

FR-ONB-006 When every task on a checklist is complete, the checklist shall move to `COMPLETED` with a completion timestamp.

FR-ONB-007 Offboarding shall be recorded as described in §6.9, carrying the relieving date, the reason, notes and a full-and-final status.

20. Performance Management

FR-PRF-001 An employee shall be able to record a performance goal carrying a title, a description, a progress percentage and a status, defaulting to active. **FR-PRF-002** An employee shall be able to list their own goals. **FR-PRF-003** The system shall support a performance review for an employee and a manager covering a review period, carrying a self rating and comment, a manager rating and comment, and a status defaulting to draft. **FR-PRF-004** An employee shall be able to list their own reviews. **FR-PRF-005** A manager shall be able to list their team's reviews.

21. Dashboards and Analytics

21.1 Personal Dashboard

FR-DSH-001 Every signed-in user shall have a personal dashboard reporting: their name and employee code; whether they have punched in today with the punch-in and punch-out times and worked minutes; their current-year leave balances; the number of their pending leave requests; the number of their tickets not yet closed; the number of assets assigned to them; and their five most recent notifications.

21.2 Executive Dashboard

FR-DSH-002 Holders of DASHBOARD_EXEC shall have an executive dashboard, optionally filtered to one division, reporting: headcount, present today, attendance percentage, pending leave approvals, open tickets, assets assigned and assets in stock, together with departmental, attendance-trend, leave-utilisation and payroll-cost breakdowns.

FR-DSH-003 Headcount shall exclude offboarded staff, so that the total and the attendance percentage agree with each other and with the administrative views.

FR-DSH-004 Present-today, pending-approval and open-ticket counts shall be restricted to records whose owner falls within the selected division, with an unset filter meaning the whole organisation.

FR-DSH-005 The attendance percentage shall be present today divided by headcount, expressed as a percentage rounded half-up to one decimal place, and shall be zero when headcount is zero.

FR-DSH-006 Division membership shall be resolved once per request into an in-memory index so that per-record filtering does not repeatedly query the database.

21.3 Celebrations

FR-DSH-007 Every signed-in user shall be able to read upcoming celebrations: birthdays and work anniversaries, optionally restricted to one division.

FR-DSH-008 Only enabled, non-offboarded employees shall be considered.

FR-DSH-009 An occurrence shall be projected onto the current year and rolled forward to the next year where it has already passed; 29 February shall be treated as 1 March.

FR-DSH-010 Occurrences more than **60 days** away shall be excluded, so that the panel shows what is coming with enough notice for an anniversary worth marking.

FR-DSH-011 An anniversary shall be reported only from the first completed year, and shall carry the number of completed years.

FR-DSH-012 Results shall be ordered by days remaining and limited to **twelve** entries, with the division filter applied **before** the limit so that narrowing to a division returns twelve of that division.

FR-DSH-013 Each entry shall carry the employee's identifier, name, employee code, team, photograph, the occurrence type, the date, the days remaining and, for anniversaries, the completed years.

21.4 Side-service Analytics

FR-DSH-014 The biometric side service shall expose per-employee analytics and organisation-wide executive analytics, the latter accepting a division filter.

22. Notifications and Alerts

FR-NOT-001 The system shall persist every notification with a recipient, a title, a body, a type, a target link and a read flag.

FR-NOT-002 Notification creation shall be asynchronous, so that no business action — a leave approval, an asset allocation, a task assignment — is ever delayed by delivery.

FR-NOT-003 Each notification shall be pushed to the recipient's user-scoped queue and to a per-user public topic, so that clients subscribing either way receive it.

FR-NOT-004 A user shall be able to browse their notifications, paged, most recent first; to read their unread count; to mark one as read; and to mark all as read.

FR-NOT-005 Marking a notification read shall apply only to the caller's own notification.

FR-NOT-006 Notification types shall be at least: LEAVE, PERMISSION, TASK, HELPDESK, CHAT, CELEBRATION, ASSET, PAYROLL, PAYSIP, CLAIM, COMPLAINT, SAFETY, CALENDAR, ANNOUNCEMENT and SYSTEM. Each shall be rendered with a distinguishing icon and colour.

FR-NOT-007 A scheduled job shall run **once daily at 09:00 server time** and shall notify every active employee of every birthday and work anniversary falling that day.

FR-NOT-008 The celebrant shall not be notified of their own day.

FR-NOT-009 The daily celebration job shall additionally send one bulk SMS to the whole active workforce excluding the celebrant.

FR-NOT-010 The web client shall subscribe to the live channel on sign-in, shall surface new notifications immediately without a page reload, and shall invalidate the affected cached queries so that the underlying screens refresh themselves.

FR-NOT-011 Notification bodies shall be human-readable and shall name the subject, the action and the relevant dates or amounts.

23. Internal Communication

23.1 Channels

FR-CHT-001 Administrators shall be able to create a community channel with a name, a description and an announcement flag; the creator shall be added as a member automatically.

FR-CHT-002 A channel name that duplicates an existing name, case-insensitively, shall be rejected; a name using the reserved direct-message prefix shall be rejected.

FR-CHT-003 Administrators shall be able to delete a channel and to add or remove members.

FR-CHT-004 Any signed-in user shall be able to list the channels available to them and the channels they belong to, and to read a channel's members.

FR-CHT-005 An organisation-wide announcement channel shall exist, and every newly created employee shall be enrolled in it automatically.

FR-CHT-006 Direct one-to-one conversations shall be backed by hidden two-member rooms named with a reserved prefix, found or created on demand, kept out of the administrative channel listing, and guaranteed to contain exactly the two participants.

FR-CHT-007 Starting a conversation with oneself shall be rejected.

FR-CHT-008 Team conversations shall be backed by private per-team rooms named with a reserved prefix derived from the team title, found or created on demand, with every teammate enrolled. A user with no team shall be told they are not yet assigned to one. Team rooms shall be kept out of the general chat listing.

FR-CHT-009 Any signed-in user shall be able to list their contacts for starting a conversation.

23.2 Messages

FR-CHT-010 A member shall be able to read a channel's messages and to post a text message. **FR-CHT-011** A member shall be able to post a voice note and file attachments. **FR-CHT-012** A member shall be able to delete a message they sent. **FR-CHT-013** New messages shall be pushed live to channel members and shall raise a notification of type CHAT. **FR-CHT-014** Chat events shall be publishable to and consumable from an event stream, so that delivery can scale beyond a single application instance; absence of the stream shall not prevent direct delivery. **FR-CHT-015** The system shall relay call signalling messages between participants to support voice or video calling. **FR-CHT-016** Attachments shall respect the configured per-file and per-request size limits, accommodating short video clips.

24. AI Assistant

FR-BOT-001 The system shall provide a conversational assistant available from every authenticated screen.

FR-BOT-002 The assistant shall report its own capability set — whether it is enabled, which languages it offers, and whether speech synthesis and transcription are available — so that the interface only offers what is actually configured.

FR-BOT-003 The assistant shall support at least English, Tamil and Hindi, and shall greet the user in the selected language by name.

FR-BOT-004 The assistant shall answer questions using a built-in guide to the system's own functionality, augmented by an administrator-curated knowledge base.

FR-BOT-005 The assistant shall retain a bounded conversation history — at most **8** turns — and shall bound the knowledge context it supplies to the model to at most **6 000** characters.

FR-BOT-006 The assistant shall be able to answer with organisation context for privileged callers, and shall withhold that context otherwise.

FR-BOT-007 The assistant shall expose a translation function.

FR-BOT-008 The assistant shall expose speech synthesis, selecting a voice and model appropriate to the requested language and enforcing the correct model when a non-English language is requested.

FR-BOT-009 The assistant shall expose speech-to-text over an uploaded audio file with an optional language hint.

FR-BOT-010 Holders of `USER_MANAGE` shall be able to read and update assistant settings: the master enable switch, the chosen language-model provider, provider keys and model names for chat, transcription and speech, the crawl provider key, and the organisation website address.

FR-BOT-011 Provider keys shall never be returned in full; any display shall be masked to a four-character prefix and suffix, or fully masked for short values.

FR-BOT-012 A blank or absent setting value shall be ignored on update, so that a key is never wiped by accident.

FR-BOT-013 Holders of `USER_MANAGE` shall be able to ingest a website into the knowledge base, bounded to at most **12 000** characters per ingestion, and shall be able to list and delete knowledge documents.

FR-BOT-014 Where a provider is unreachable or unconfigured, the assistant shall degrade gracefully with an explanatory reply rather than failing the request.

25. Reports and Data Export

FR-RPT-001 Holders of `REPORT_VIEW` shall be able to export an attendance report for a date range with an optional department filter, as a spreadsheet, delivered as a file attachment with a meaningful filename.

FR-RPT-002 Holders of `REPORT_VIEW` shall be able to export a leave report for a date range with an optional department filter, as a spreadsheet attachment.

FR-RPT-003 Holders of `REPORT_VIEW` shall be able to export a payroll report for a month and a year, as a spreadsheet attachment.

FR-RPT-004 Task and work-report exports shall be provided as specified in §12.4 and §13.

FR-RPT-005 Every export shall be generated in memory and streamed; no temporary file shall be left behind.

FR-RPT-006 Every export shall carry a header row and auto-sized columns.

26. Calendar and Holidays

FR-CAL-001 Any signed-in user shall be able to read the holiday calendar, optionally for a single year, ordered by date ascending. **FR-CAL-002** Holders of `ORG_MANAGE` or `CALENDAR_MANAGE` shall be able to add a holiday with a name and a

date, and to delete one. **FR-CAL-003** Adding a holiday shall notify **every** active, non-offboarded employee in the feed, linking to the calendar, and shall send one bulk SMS naming the entry and its date. **FR-CAL-004** Holidays shall be excluded from working-day counts everywhere they are computed — leave working days, monthly attendance summaries, loss-of-pay day counts and loss-of-pay previews. **FR-CAL-005** The system shall be seeded with a national and regional holiday calendar so that the calendar screen carries real content from first run, each entry inserted only when the same date and name are absent. **FR-CAL-006** The calendar screen shall present holidays alongside approved leave, so that absence is visible in context.

27. File Storage and Documents

FR-FIL-001 The system shall store uploaded files under a configured storage root, creating it at startup where absent. **FR-FIL-002** An uploaded file shall be stored under a folder-and-month path with a generated unique name preserving the original extension, and the relative path shall be returned to the caller. **FR-FIL-003** An empty upload shall be rejected. **FR-FIL-004** Generated artefacts — payslip PDFs and QR images — shall be written by the same mechanism under their own folders. **FR-FIL-005** Stored files shall be readable by relative path; a missing file shall report not found. **FR-FIL-006** File paths shall be normalised so that a request cannot escape the storage root. **FR-FIL-007** File retrieval shall be publicly reachable so that images and documents render without a bearer token, and shall serve content as an opaque byte stream that is never executed or interpreted. **FR-FIL-008** The storage mechanism shall be swappable — local disk, or an object store — behind a single abstraction, selected by configuration, without changes to callers. **FR-FIL-009** Employee documents shall be recorded on the employee as a comma-separated list of stored paths, the same shape attachments take elsewhere in the system.

28. System Settings

FR-SET-001 The system shall hold configurable settings as key-value pairs with a description and an update timestamp. **FR-SET-002** Any signed-in user shall be able to read the settings map. **FR-SET-003** Holders of `USER_MANAGE` shall be able to update settings. **FR-SET-004** Settings shall include at least: the assistant enable switch, the assistant language-model provider, provider keys and model names for chat, transcription and speech synthesis, the crawl provider key, the organisation website address, and the per-kilometre travel rates for hills and plains. **FR-SET-005** Settings holding secrets shall never be returned in full to a client.

29. Mobile Application

FR-MOB-001 The mobile application shall support the four things employees most need away from a desk: their dashboard, attendance punching, leave, and payslips. **FR-MOB-002** The mobile application shall authenticate against the same API and shall store credentials in the platform's secure store. **FR-MOB-003** The mobile application shall present tabbed navigation across Home, Attendance, Leave and Payslips, and shall offer sign-out. **FR-MOB-004** The mobile application shall obtain the device's location for a punch and shall submit it with the punch. **FR-MOB-005** The mobile application shall present the signed-in employee's dashboard, leave balances and history, and payslip list with download. **FR-MOB-006** The mobile application shall be built from a single codebase for both mobile platforms. **FR-MOB-007** The mobile application shall show the sign-in screen whenever no authenticated session exists.

30. Biometric and Document Intelligence Service

FR-BIO-001 The side service shall expose a health or root endpoint. **FR-BIO-002** The side service shall enrol a face image against an employee identifier and shall persist the enrolment. **FR-BIO-003** The side service shall verify a submitted face image against an employee's enrolment and shall report the match outcome. **FR-BIO-004** The side service shall extract numeric text from an uploaded image. **FR-BIO-005** The side service shall expose per-employee and executive analytics as described in §21.4. **FR-BIO-006** The side service shall initialise its recognition models once at startup rather than per request. **FR-BIO-007** The side service shall restrict cross-origin access to the configured web origins. **FR-BIO-008** The side service shall be optional; the platform shall function fully without it, with dependent

features reporting themselves unavailable. **FR-BIO-009** The side service shall not be publicly reachable; access shall be routed through the single public entry point.

Part III — Data, Rules and Quality

31. Data Requirements

31.1 Logical Entity Inventory

DR-004 The system shall hold the following persistent entities, grouped by subject area:

Area	Entities
Identity & security	users, roles, permissions, role_permissions, user_roles, refresh_tokens, login_history, audit_log, otp_codes
Employee detail	bank_details, employee_documents, family_members, educations, experiences, worker_skills, offboarding_records
Organisation	companies, branches, departments, designations, positions, employment_statuses, blood_groups, office_locations, shifts, sites, holidays, system_settings
Attendance	attendance
Leave	leave_types, leave_balances, leave_requests, permission_requests
Payroll	salary_structures, payroll_runs, payslips, payslip_requests, investment_declarations
Tasks & work	tasks, work_reports
Expenses	ta_expenses, expense_claims
Assets	assets, asset_allocations, asset_maintenance, software_licenses
Service management	tickets, ticket_comments, complaints_needs, safety_incidents
Lifecycle	onboarding_checklists, onboarding_tasks
Performance	performance_goals, performance_reviews
Engagement	notifications, announcements, communities, community_members, community_messages
AI	chatbot_knowledge

31.2 Common Column Conventions

DR-005 Every business entity shall carry a surrogate auto-increment primary key. **DR-006** Auditable entities shall carry created_by, updated_by, created_at and updated_at, populated automatically. **DR-007** Text storage shall use utf8mb4, so that Indian-language content and emoji are stored without loss. **DR-008** Monetary columns shall be DECIMAL(12,2); day quantities DECIMAL(5,1); hour quantities DECIMAL(4,1) or DECIMAL(4,2); coordinates DECIMAL(10,7); percentages DECIMAL(5,2). **DR-009** Status and type columns shall be short variable-length strings holding upper-case codes, with the permitted set enforced in application logic. **DR-010** Every foreign key shall declare its deletion behaviour; where a reference is intentionally weak, the application shall clear it explicitly on deletion.

31.3 Key Entity Definitions

`users` — the employee record and the identity.

Group	Columns
Identity	id, employee_code (unique), username (unique), name, dob, gender, aadhar (unique), phone (unique), email, personal_email, photo_path, documents
Security	password_hash, password_vault, enabled, failed_login_count, locked_until, last_login_at
Address	care_of, house, street, locality, vtc, district, state, country, pincode, post_office
Statutory	pan, pf_number, blood_group
Contact	alternate_phone, emergency_contact, emergency_contact_relation
Employment	blood_group_id, department_id, designation_id, office_location_id, employment_status_id, position_id, reporting_manager_id, site_id, employment_type, date_of_joining, industry
Free-text titles	designation_title, department_title, position_title, tech_stack
Lifecycle	profile_status
Audit	created_by, updated_by, created_at, updated_at

Indexed on department and reporting manager.

`attendance` — one row per employee per day, uniquely constrained on employee and work date so a duplicate punch-in cannot be recorded. Holds punch-in and punch-out timestamps, mode, in and out coordinates, site and shift references, geofence result and exception flag, status, late flag and late minutes, worked minutes and overtime minutes. Indexed on work date.

`leave\_types` — name, unique code, maximum days per year, carry-forward, encashable, gender restriction, past-dates allowance, accrual type, minimum notice days, per-period limit, paid flag, active flag.

`leave\_balances` — employee, leave type, year, allocated and used, uniquely constrained on the triple.

`leave\_requests` — employee, leave type, from and to dates, working days, reason, attachment path, status, named approver, decider, decision timestamp and comment. Indexed on status and employee.

`permission\_requests` — employee, request date, from and to times as HH:mm, hours, reason, status, named approver, decider, decision timestamp and comment.

`salary\_structures` — employee, basic salary, house-rent allowance, allowances, provident-fund amount, ESI applicability, professional tax, effective-from date, active flag.

`payroll\_runs` — pay month and year uniquely constrained, status, initiator, run timestamp, finance approver and approval timestamp.

`payslips` — payroll run reference, employee, pay month and year uniquely constrained together with the employee; the earnings members basic, house-rent allowance, allowances, overtime, performance and expenses; the deduction members provident fund, ESI, professional tax, tax deducted at source, health insurance, salary advance and other; gross, total deductions and net pay; loss-of-pay days; working days; pay date; the company overrides for name, logo, tax registration number and address; bank name and account; designation and department; the source marker; the generated PDF path; and the generation timestamp.

`payslip\_requests` — employee, pay month and year, note, status, generated payslip reference, decider, decision timestamp and note.

`tasks` — title, description, assignee, assigner, status, priority, progress, due date, completion timestamp, team batch identifier and team name.

`work\_reports` — employee, work date, project name, work hours, task description.

`ta\_expenses` — employee, date, location, category, starting and ending odometer readings, total, hills and plains kilometres, total amount, bus fare, other amounts, gross total, remarks, status, fuel-receipt path, attachment paths, decision comment, decider and decision timestamp.

`assets` — unique asset code, category, asset type, brand, model, serial number, registration number, purchase date and cost, warranty and maintenance-contract expiry, status, quantity, site reference, assignee, QR path, depreciation rate.

`asset\_allocations` — asset, employee, allocation timestamp, acknowledgement flag and timestamp, return timestamp and condition.

`tickets` — unique ticket code, raiser, title, description, attachments, type, category, priority, status, assignee, SLA due time, rating, resolution timestamp.

`complaints\_needs` — unique reference code, raiser, kind, category, subject, description, priority, status, HR response, named recipient, handler, resolution timestamp.

`safety\_incidents` — unique reference code, reporter, site, incident type, description, zone, anonymity flag, severity, status, occurrence timestamp, resolution notes, resolver and resolution timestamp.

`notifications` — recipient, title, body, type, link, read flag, creation timestamp.

`communities`, **`community\_members`**, **`community\_messages`** — channel name, description, creator, announcement flag; the membership pair with a join timestamp; and messages with sender, content, attachments, voice path and send timestamp.

31.4 Reference Data and Seeds

DR-011 The system shall seed, on first run: the role catalogue, the permission catalogue and the role-permission grants; blood groups; departments; designations; employment statuses; positions; office locations with coordinates and geofence radii; shifts; project sites; the leave types with their entitlements and restrictions; a holiday calendar; announcements; and a set of demonstration accounts.

DR-012 Seeded and administrative accounts shall exist for the board-level administrators, and the designated escalation approver shall hold administrative access.

DR-013 Every seed shall be idempotent, inserting only where the row is absent.

DR-014 Migrations shall be provided that clear transactional demonstration data for a fresh start without disturbing master data.

31.5 Audit, Retention and Integrity

DR-015 Every sign-in attempt shall be retained with its outcome, identifier, IP address and user agent. **DR-016** A general audit log shall be available recording the actor, the action, the entity type and identifier, free-text detail, the IP address and the timestamp, indexed on actor and on entity. **DR-017** Every auditable entity shall retain who created and last updated it, and when. **DR-018** Decisions shall retain the decider, the decision timestamp and the decision comment, so that every approval, rejection and status change is attributable. **DR-019** Soft deletion shall be used where history matters — notably leave types — and hard deletion only where a record is genuinely to be expunged. **DR-020** Generated documents shall be retained with their stored path so that a historical payslip remains retrievable. **DR-021** Sequence-derived codes shall be generated by incrementing the observed maximum for the period, never by counting rows, so that deletion cannot cause a code to be reissued.

31.6 Schema Evolution

DR-022 All schema change shall be delivered as numbered, forward-only migrations applied automatically at startup.

DR-023 Migrations shall be permitted to baseline an existing database and to repair their own metadata, and shall not fail startup on checksum drift. **DR-024** Where a migration must reconcile a live schema with the object model, it shall check the schema catalogue before each change so that it is safe on both an empty and a populated database. **DR-025**

The object model shall be validated against the schema at startup; a mismatch shall prevent the application from starting rather than silently corrupting data.

32. Consolidated Business Rules

Calendar and time

ID	Rule
BR-001	Sunday is the single weekly off; Saturday is a working day.
BR-002	Working days exclude Sundays and configured holidays.
BR-003	Office hours are 09:00–18:00 unless the assigned shift overrides the start.
BR-004	The lateness grace period is zero minutes: at or before the office start is on time; after it is late by the difference.
BR-005	Overtime accrues only past the office end; a person starting after the office end earns only from when they actually began.
BR-006	All recorded times are in the organisation's local time zone.

Attendance

ID	Rule
BR-007	One punch-in and one punch-out per employee per day; duplicates are refused.
BR-008	A punch-out requires an existing punch-in.
BR-009	Work-from-home punches are not geofence-evaluated.
BR-010	Absence of coordinates never blocks a punch; it is recorded outside the geofence with an exception flag.
BR-011	The default geofence radius is 200 metres where the location defines none.
BR-012	Monthly working days are counted only up to today within the current month.
BR-013	Absentees are computed from actual punch and leave data regardless of weekday.

Leave

ID	Rule
BR-014	Gender-restricted leave types are refused to a mismatched profile.
BR-015	Past-dated leave is refused unless the type explicitly allows it.
BR-016	Minimum notice applies only to future-dated leave.
BR-017	A range containing no working days is refused.
BR-018	Casual and sick leave are limited to one per calendar quarter, expressed through the type's per-period limit.

ID	Rule
BR-019	Every type except Loss of Pay requires an allocated balance and sufficient availability, checked both at application and again at approval.
BR-020	Balance is consumed on approval, not on application.
BR-021	Cancelling an approved leave refunds the balance, floored at zero.
BR-022	Only pending or approved leave may be cancelled, and only by its owner.
BR-023	A rejection requires a reason.
BR-024	Bulk allocation never overwrites an existing balance and never allocates for a past year.
BR-025	Leave routing is single-level and duration-dependent: up to three days to the applicant's own Team Leader, beyond three days to a Manager; a Team Leader's leave to a Manager; HR's leave to one named escalation approver.
BR-026	Nobody may decide their own leave.
BR-027	Where a specific approver was named, only that person may act.

Hourly permissions

ID	Rule
BR-028	The end time must be after the start time.
BR-029	Duration is computed to two decimal places from the minute difference.
BR-030	A pending request whose date has passed becomes overdue and can no longer be decided.
BR-031	Routing is single-level: employee to own-team Team Leader, Team Leader to HR, HR to the named escalation approver.
BR-032	Only pending requests may be cancelled, and only by their owner.

Payroll

ID	Rule
BR-033	No payslip may exist for a month before the employee's joining month.
BR-034	Payslip generation requires an active salary structure.
BR-035	At most one payslip per employee per month.
BR-036	The provident-fund figure on a salary structure is a flat rupee amount.
BR-037	ESI is 0.75 % of gross, applied only where the structure marks it applicable and gross does not exceed the wage ceiling of 21 000.

ID	Rule
BR-038	Overtime pay uses an hourly rate of gross monthly pay divided by 240.
BR-039	Automatic attendance-driven loss of pay is not deducted; loss of pay is entered manually so the two cannot double-count.
BR-040	The loss-of-pay day count is still computed from attendance for reporting, excluding Sundays, holidays and future days.
BR-041	Net pay is gross minus total deductions; salary advance is a deduction line.
BR-042	Monetary results are scaled to two places with half-up rounding.
BR-043	A payslip may only be read or downloaded by its owner, or by a privileged caller.
BR-044	A payslip PDF is regenerated at download so corrected metadata is always reflected.
BR-045	At most one payroll run per month.
BR-046	The run workflow is preview → confirmed → finance-approved → paid, and each transition validates the current state.
BR-047	A batch run skips employees for whom generation is not possible rather than failing.
BR-048	At most one pending payslip request per employee per month.
BR-049	An approved payslip request stores exactly the values the administrator entered, so the download reproduces that format.

Tasks

ID	Rule
BR-050	HR may assign only to Team Leaders; a Team Leader may assign only within their own team; a full administrator is unrestricted.
BR-051	A due date may not be in the past.
BR-052	Task state moves forward only: pending → in progress → completed, never back.
BR-053	Progress and status stay consistent: zero is pending, 100 is completed, anything between is in progress.
BR-054	Only the assignee may report progress or complete a task.
BR-055	Only the assigner, or an administrator, may edit a task; progress and status remain the assignee's.
BR-056	A Team Leader sees only their own team's tasks.

Claims

ID	Rule
----	------

ID	Rule
BR-057	An owner may correct their claim only while it is pending; an approver may correct any claim at any time.
BR-058	A rejection requires a reason.
BR-059	A Team Leader sees only their own team's claims.
BR-060	An owner may delete their own claim; an administrator may delete any.

Assets

ID	Rule
BR-061	An out-of-stock, exhausted, retired or lost asset cannot be allocated.
BR-062	Allocation decrements quantity and marks the asset out of stock at zero.
BR-063	Only the assignee may acknowledge receipt.
BR-064	A return increments quantity and sets status from the returned condition.
BR-065	Deleting an asset removes its allocation history first.

Helpdesk

ID	Rule
BR-066	Ticket category is derived from the raiser's division, never chosen.
BR-067	The SLA due time follows the priority: 4, 8, 24 or 48 hours.
BR-068	A raiser may correct their ticket only while it is open.
BR-069	A raiser may never decide their own ticket.
BR-070	Where a ticket names a recipient, only that recipient may progress it.
BR-071	Status moves forward one step at a time, except that in-progress may go to awaiting-parts or straight to resolved.
BR-072	Only the raiser may rate, and only once resolved or closed.

Complaints, safety and general

ID	Rule
BR-073	Unrecognised enumerated values fall back to a safe default rather than being rejected, for kind, priority, incident type and severity.
BR-074	An anonymous safety report never reveals its reporter.
BR-075	Reaching a terminal state sets the resolution timestamp; returning to an earlier state clears it.
BR-076	The submitter is never notified of their own submission.

ID	Rule
BR-077	Notification and SMS failure never blocks the originating business action.
BR-078	Every newly created employee joins the organisation-wide announcement channel.
BR-079	Display labels for divisions are canonicalised on write so they never leak into stored data.
BR-080	Blank optional strings are stored as absent, never as empty text.

33. Non-Functional Requirements

33.1 Performance

NFR-PERF-001 An interactive read endpoint shall respond within **500 ms** at the 95th percentile under expected load, excluding network latency. **NFR-PERF-002** A write endpoint shall respond within **1 s** at the 95th percentile, excluding asynchronous side effects. **NFR-PERF-003** Notification delivery, SMS dispatch and other side effects shall be asynchronous and shall not contribute to request latency. **NFR-PERF-004** Every list endpoint returning potentially unbounded data shall be paginated, with a default page size of 20. **NFR-PERF-005** Aggregations shall resolve related entities in batch rather than per row; per-record loops shall not issue a query each. **NFR-PERF-006** Lazy-loading outside a transaction shall be disabled, so that view rendering cannot trigger unbounded queries. **NFR-PERF-007** The database connection pool shall be bounded, with a maximum of 10 connections and a minimum of 2 idle, and shall be named for observability. **NFR-PERF-008** Indexes shall exist on every column used for routine filtering: work date, request status, employee references, department, reporting manager, audit actor and audit entity. **NFR-PERF-009** Spreadsheet exports shall be generated in memory and streamed as a single response. **NFR-PERF-010** PDF fonts and logos shall be loaded once at class initialisation, not per document. **NFR-PERF-011** The web client shall cache server state and shall refetch on invalidation rather than polling. **NFR-PERF-012** The web application shall be built as an optimised production bundle with code splitting.

33.2 Scalability and Capacity

NFR-SCAL-001 The system shall support at least **1 000** employee records and **200** concurrent signed-in users on a single application instance. **NFR-SCAL-002** The API shall be stateless, so that instances can be added behind a load balancer without session affinity. **NFR-SCAL-003** Live message distribution shall be able to use an external event stream so that delivery scales beyond one instance. **NFR-SCAL-004** File storage shall be swappable for an object store without code change, so that storage capacity is not bounded by the application host. **NFR-SCAL-005** The application heap shall be explicitly bounded in production.

33.3 Security

NFR-SEC-001 All external traffic shall use TLS in production. **NFR-SEC-002** Passwords shall be hashed with BCrypt; the hash shall be the sole authority for authentication. **NFR-SEC-003** The reversible password copy shall be encrypted with AES-GCM under a key derived from the application signing secret, so that a database dump alone — including a pre-deployment backup — carries nothing readable. Anyone holding both the dump and the application secret can read every password; this is the accepted cost of the reveal capability and shall be documented to the operator. **NFR-SEC-004** The token signing secret shall be at least 32 random characters, shall be supplied only by environment variable outside development, and shall have no fallback default. **NFR-SEC-005** Access tokens shall be short-lived and refresh tokens shall be rotated on every use and revoked wholesale on sign-out and password change. **NFR-SEC-006** Authorisation shall be enforced server-side at every endpoint; client-side hiding shall never be the only control. **NFR-SEC-007** Service-layer scoping shall additionally restrict result sets to what the caller may see, so that a broad endpoint guard cannot leak another team's or another division's data. **NFR-SEC-008** Cross-origin access shall be restricted to an

explicit allow-list. **NFR-SEC-009** Cross-site request forgery protection shall be disabled only because the API is stateless and token-based; no cookie-borne credential shall be introduced without reinstating it. **NFR-SEC-010** Input shall be validated declaratively at the boundary, and validation failures reported per field. **NFR-SEC-011** Database access shall use parameter binding. Where a native statement must name a table or column, those names shall be fixed constants and never derived from user input. **NFR-SEC-012** Stored file paths shall be normalised against the storage root so that traversal outside it is impossible. **NFR-SEC-013** Uploaded content shall never be executed or interpreted. **NFR-SEC-014** Upload sizes shall be bounded at both the application and the reverse proxy. **NFR-SEC-015** Error responses shall not disclose stack traces, SQL, internal paths or configuration. **NFR-SEC-016** Failed authentication shall not reveal whether the identifier exists; the same message shall be returned for an unknown user and a wrong password. **NFR-SEC-017** Accounts shall lock after repeated failures to frustrate credential stuffing. **NFR-SEC-018** Secrets shall never be committed to source control, never logged in full, and never returned to a client; any display shall be masked. **NFR-SEC-019** Only the health endpoint shall be exposed among operational endpoints, and it shall not disclose detail. **NFR-SEC-020** Internal services — database, application, side service — shall be unreachable from outside the deployment network. **NFR-SEC-021** Sensitive reads — notably the password reveal — shall be separate, explicitly invoked endpoints so that the value never travels with an ordinary read, a listing or an export. **NFR-SEC-022** Security-relevant events shall be logged: lockouts, sign-in attempts, and decision actions with their actor. **NFR-SEC-023** The framework security log level shall be restrained in production so that logs do not accumulate sensitive detail.

33.4 Reliability and Availability

NFR-REL-001 The system shall target **99.5 %** monthly availability during business hours. **NFR-REL-002** Optional dependencies — cache, mail, event stream, side service, SMS gateway, AI providers — shall not affect reported health and shall not prevent core operation. **NFR-REL-003** A failure in a side effect shall never fail the originating transaction. **NFR-REL-004** Business operations shall be transactional; a partial write shall not be observable. **NFR-REL-005** Batch operations shall isolate each item so that one failure does not roll back the batch. **NFR-REL-006** Containers shall restart automatically unless deliberately stopped. **NFR-REL-007** The database container shall report readiness through a health check before dependants start. **NFR-REL-008** A deployment shall verify health after starting and shall roll back automatically on failure. **NFR-REL-009** Uniqueness constraints shall prevent duplicate punches, duplicate payslips, duplicate balances and duplicate runs at the database level, not merely in application logic.

33.5 Usability and Accessibility

NFR-USE-001 The interface shall be usable from a 320 px viewport upwards without horizontal scrolling of the page body. **NFR-USE-002** Wide content — tables, wide grids — shall scroll within its own container. **NFR-USE-003** Every action shall give immediate feedback: a loading state, then a success or failure message carrying the server's own wording. **NFR-USE-004** Error messages shall be written for the person reading them, naming what went wrong and what to do, not internal codes. **NFR-USE-005** Interactive controls shall be reachable and operable by keyboard, and shall carry accessible labels where their purpose is conveyed only by an icon. **NFR-USE-006** Colour shall not be the sole carrier of meaning; status shall also be conveyed by text. **NFR-USE-007** Text and interface contrast shall meet WCAG 2.1 AA in both light and dark themes. **NFR-USE-008** The chosen theme shall persist across sessions. **NFR-USE-009** Dates shall be presented consistently and unambiguously throughout. **NFR-USE-010** Destructive actions shall require explicit confirmation. **NFR-USE-011** Navigation shall never present a route the user cannot use. **NFR-USE-012** Role names shown to users shall be the friendly business names, not internal codes.

33.6 Compatibility and Portability

NFR-COMP-001 The web application shall function on current versions of the major browsers. **NFR-COMP-002** The mobile application shall function on currently supported Android and iOS versions. **NFR-COMP-003** The API shall preserve the legacy contract shapes it deliberately mirrors — dropdown by type, bank action semantics, payslip components — so that existing integrations have direct equivalents. **NFR-COMP-004** The server side shall run wherever a compliant Java runtime and MySQL are available, and shall be deployable as containers. **NFR-COMP-005** Configuration shall be supplied entirely by environment variables and database-held settings, so that the same image serves every environment. **NFR-COMP-006** No feature shall depend on a specific host operating system.

33.7 Maintainability

NFR-MAIN-001 Each functional area shall be a self-contained module comprising entity, repository, service, controller and data-transfer objects, with cross-cutting concerns held outside the modules. **NFR-MAIN-002** Adding a module shall follow a documented, repeatable recipe requiring no architectural decision. **NFR-MAIN-003** Business logic shall live in the service layer; controllers shall handle transport and authorisation only. **NFR-MAIN-004** Data-transfer objects shall be immutable records carrying declarative validation. **NFR-MAIN-005** Configuration shall be bound to strongly-typed objects rather than read as loose strings. **NFR-MAIN-006** Every non-obvious decision shall be documented at the point of code that implements it, stating the reason. **NFR-MAIN-007** External integrations shall sit behind an abstraction so that a provider can be replaced without touching callers. **NFR-MAIN-008** Documentation-only changes shall not trigger a production rebuild.

33.8 Localisation and Internationalisation

NFR-LOC-001 All storage shall be utf8mb4, so that Indian-language text and emoji round-trip without loss. **NFR-LOC-002** The AI assistant shall converse in at least English, Tamil and Hindi. **NFR-LOC-003** Generated documents shall embed fonts capable of rendering the currency symbol and Indian-language text. **NFR-LOC-004** The system shall operate in the organisation's local time zone consistently across containers, so that a recorded punch time is the wall-clock time the employee saw. **NFR-LOC-005** Monetary presentation shall use the Indian rupee. **NFR-LOC-006** A translation resource shall be maintained for regional-language interface text.

33.9 Observability

NFR-OBS-001 The application shall log at information level for its own packages and shall restrain framework logging. **NFR-OBS-002** Every outbound integration attempt shall log its outcome, and a failure shall log enough to diagnose it without disclosing a secret. **NFR-OBS-003** The active SMS provider and its configuration state shall be logged at startup, with the key masked, and a misconfiguration — enabled without a key — shall be logged as a warning. **NFR-OBS-004** Unhandled exceptions shall be logged with their stack trace server-side while returning a generic message to the client. **NFR-OBS-005** A health endpoint shall be available for uptime monitoring. **NFR-OBS-006** Deployment scripts shall log each step and shall fail loudly rather than continuing after an error.

33.10 Backup and Recovery

NFR-BAK-001 A database backup shall be taken automatically before every deployment. **NFR-BAK-002** Backups shall be retained outside the application working directory. **NFR-BAK-003** A documented rollback procedure shall restore both the previous application version and, where required, the pre-deployment database state. **NFR-BAK-004** Database and file storage shall be held in durable volumes that survive container replacement. **NFR-BAK-005** Recovery point objective shall be one day; recovery time objective shall be four hours.

33.11 Compliance and Privacy

NFR-PRIV-001 Personal data — national identifier, PAN, provident-fund number, bank details, date of birth, address, emergency contact — shall be readable only by the employee themselves and by roles holding an explicit employee-management or payroll capability. **NFR-PRIV-002** Bank details shall be readable by the owner and by payroll and employee-management roles only. **NFR-PRIV-003** Safety reports marked anonymous shall never disclose their reporter, in any view, notification or export. **NFR-PRIV-004** Every access to and change of employee data shall be attributable through the audit trail. **NFR-PRIV-005** Statutory computations shall be configurable so that a change in rate or ceiling requires configuration, not code. **NFR-PRIV-006** The reversible-password capability shall be documented as a deliberate, security-relevant trade-off, with its blast radius stated, so the operator can make an informed decision about retaining it.

34. Deployment and Operations

34.1 Environments

OR-001 The system shall support at least three environments — development, staging and production — differentiated only by configuration. **OR-002** The development profile shall supply its own fallback signing secret so that a local run works unconfigured; no other profile shall. **OR-003** Production shall run as a container composition on a single host: database, application, optional side service, and a reverse proxy that is the only public entry point. **OR-004** The database shall not publish a host port; only containers on the internal network shall reach it.

34.2 Configuration

OR-005 The following shall be configurable by environment variable:

Variable	Purpose
DB_HOST, DB_PORT, DB_NAME, DB_USER, DB_PASSWORD	Database connection
APP_JWT_SECRET	Token signing secret — mandatory outside development
ACCESS_TOKEN_TTL_SECONDS, REFRESH_TOKEN_TTL_SECONDS	Session lifetimes
APP_CORS_ALLOWED_ORIGINS	Allowed web origins
STORAGE_PATH	File storage root
OFFICE_START, OFFICE_END	Office hours
REDIS_HOST, REDIS_PORT	Cache
KAFKA_HOST, KAFKA_PORT	Event stream
MAIL_HOST, MAIL_PORT, MAIL_USER, MAIL_PASSWORD	Mail relay
FAST2SMS_ENABLED, FAST2SMS_API_KEY, FAST2SMS_ROUTE, FAST2SMS_SENDER_ID	Primary SMS gateway
TWILIO_ENABLED, TWILIO_ACCOUNT_SID, TWILIO_AUTH_TOKEN, TWILIO_FROM_NUMBER, TWILIO_DEFAULT_CC	Secondary SMS gateway
PORT / SERVER_PORT	Listen port, with a platform-injected value taking precedence
TZ, JAVA_OPTS	Time zone and heap bounds

OR-006 Secrets shall be supplied from a local environment file that is never committed, populated from a secrets store. **OR-007** An example environment file listing every required variable shall be maintained alongside the composition. **OR-008** Assistant provider credentials and travel rates shall be held as database settings, administrable through the interface rather than requiring a redeployment.

34.3 Continuous Deployment

OR-009 A push to the main branch shall trigger an automated deployment, and a manual trigger shall additionally offer rollback and optional service profiles. **OR-010** The pipeline shall hold no privileges beyond reading the repository, and shall verify the target host key before connecting. **OR-011** The deployment logic shall live on the server as scripts, so that the same code path can be run by hand during an incident. **OR-012** Deployment shall: back up the database; fetch and reset to the target revision; build; start; verify health; and roll back automatically on failure. **OR-013** Documentation-only changes shall be excluded from triggering a rebuild. **OR-014** Schema migrations shall be applied automatically at application startup, before the application begins serving.

34.4 Operations

OR-015 Health shall be verifiable by a single request that reports database reachability. **OR-016** Containers that record or render timestamps shall run in the organisation's local time zone. **OR-017** The reverse proxy shall permit the same upload sizes as the application. **OR-018** Database and file-storage volumes shall be durable across container replacement. **OR-019** Runbooks shall be maintained for deployment, rollback, health verification and backup restoration. **OR-020** A documented procedure shall exist for clearing transactional data for a fresh start without disturbing master data.

35. Constraints, Assumptions and Known Limitations

35.1 Known Limitations

ID	Limitation	Impact	Suggested resolution
LM-001	The reversible password copy allows any holder of both a database dump and the application signing secret to read every password.	High security exposure by design.	Review whether the reveal capability is required; if not, drop the column and the endpoint.
LM-002	Several executive-dashboard panels — departmental breakdown, monthly attendance trend, leave utilisation and payroll costs — return fixed placeholder values rather than computed figures.	The executive view is partly non-representative.	Replace each with a query over live data.
LM-003	Sign-in accepts a full-name match when the identifier is not a known username.	Weakens the uniqueness of the credential and is surprising.	Restrict sign-in to username only, or gate the fallback behind configuration.
LM-004	An administrative password reset enforces a minimum of only four characters, weaker than the policy applied elsewhere.	Weak credentials can be set administratively.	Apply one password policy to every path that sets a password.
LM-005	Approval routing for leave, permissions, tickets and complaints escalates to a hard-coded employee code for the HR level.	The escalation approver cannot be changed without a code change.	Move the escalation approver to a system setting or a role.
LM-006	The role code <code>IT_MGR</code> denotes HR and is relabelled in the interface; <code>IT_HR</code> is displayed as HR Head.	Persistent source of confusion between code and meaning.	Rename the roles in data with a migration and remove the display shim.
LM-007	Face enrolment and verification components exist but are not wired into any screen.	The capability is unavailable to users.	Wire the dialogs into the attendance and employee screens, or remove them.
LM-008	A start-up routine copies a fixed image between two absolute developer paths.	Environment-specific code in a service; no effect in production but wrong in principle.	Remove the routine and ship the asset with the web application.

ID	Limitation	Impact	Suggested resolution
LM-009	Recruitment, learning management, and several civil-specific site-operations areas are modelled but not implemented end-to-end.	Those stories cannot be completed in the product.	Build out following the documented module recipe.
LM-010	Automated test coverage is not established in the repository.	Regressions are detectable only by inspection.	Introduce a test suite covering the business rules of §32 first.
LM-011	Minimum-notice calculation approximates a month as thirty days.	Notice checks can be off by a day or two across month boundaries.	Compute the notice as a plain day difference.
LM-012	Some code sequences derive from a row count rather than the observed maximum.	A deletion can cause a code to be reissued.	Apply the maximum-plus-one approach used for complaint codes everywhere.
LM-013	Migration validation is disabled and repair enabled at startup.	Checksum drift passes silently.	Re-enable validation once the migration history is stable.
LM-014	Refresh tokens are persisted indefinitely and revoked rather than pruned.	The table grows without bound.	Add a scheduled purge of expired and revoked tokens.
LM-015	Full offline operation is not supported; only an installable shell is provided.	Field staff without connectivity cannot punch.	Add queued offline punches with deferred synchronisation.

35.2 Explicit Non-Goals

CN-012 The system shall not attempt multi-tenant isolation; one deployment serves one organisation. **CN-013** The system shall not integrate with biometric attendance hardware; the **BIOMETRIC** mode denotes a punch source, not a device protocol. **CN-014** The system shall not perform statutory filing or return submission. **CN-015** The system shall not act as a general ledger or accounting system.

36. Traceability Matrix

#	Capability	Requirements	Principal API surface	Web route	Primary data
1	Authentication and session	FR-AUTH-001...028	/api/auth/**	/login	users, refresh_tokens, login_history
2	Employee management	FR-EMP-001...053	/api/users/**, /api/auth/employees, /api/auth/employees/bulk	/employees, /profile	users, bank_details, offboarding_records
3	Master data	FR-ORG-001...011	/api/org/**	master-data pickers	departments, designations, positions, shifts, sites, office_locations, blood_groups, employment_statuses
4	Attendance	FR-ATT-001...031	/api/attendance/**	/attendance, /team-attendance	attendance

#	Capability	Requirements	Principal API surface	Web route	Primary data
5	Leave	FR-LV-001...052	/api/leave/**	/leave, /leave/approvals, /leave/policies	leave_types, leave_balances, leave_requests
6	Hourly permissions	FR-PRM-001...014	/api/leave/permissions/**	/leave/permissions	permission_requests
7	Payroll	FR-PAY-001...044	/api/payroll/**	/payslips, /payroll/requests, /payroll/run	salary_structures, payslips, payslip_requests, payroll_runs
8	Tasks	FR-TSK-001...025	/api/tasks/**	/tasks	tasks
9	Work reports	FR-WRP-001...008	/api/work-reports/**	/work-reports	work_reports
10	Travel and expense claims	FR-CLM-001...020	/api/ta-expenses/**, /api/finance/expenses/**	/ta-expenses, /ta-expenses/new	ta_expenses, expense_claims
11	Assets	FR-AST-001...017	/api/assets/**	/assets	assets, asset_allocations
12	Helpdesk	FR-HD-001...025	/api/tickets/**	/helpdesk, /helpdesk/new	tickets, ticket_comments
13	Complaints and needs	FR-CMP-001...014	/api/complaints/**	/complaints	complaints_needs
14	Safety incidents	FR-SAF-001...011	/api/safety-incidents/**	safety screens	safety_incidents
15	Onboarding	FR-ONB-001...007	/api/onboarding/**	/onboarding	onboarding_checklists, onboarding_tasks
16	Performance	FR-PRF-001...005	/api/performance/**	performance screens	performance_goals, performance_reviews
17	Dashboards	FR-DSH-001...014	/api/dashboard/**	/	aggregate across modules
18	Notifications	FR-NOT-001...011	/api/notifications/**, /ws	/notifications	notifications
19	Communication	FR-CHT-001...016	/api/communities/**, /api/calls/**	/chat, /communities	communities, community_members, community_messages
20	AI assistant	FR-BOT-001...014	/api/chatbot/**	/admin/ai-assistant, global widget	chatbot_knowledge, system_settings
21	Reports	FR-RPT-001...006	/api/reports/**	/reports	aggregate across modules
22	Calendar	FR-CAL-001...006	/api/org/holidays	/calendar	holidays
23	Files	FR-FIL-001...009	/api/files/**, upload endpoints	inline	storage volume
24	Settings	FR-SET-001...005	/api/settings	/admin/ai-assistant	system_settings
25	Mobile	FR-MOB-001...007	shared API	mobile tabs	as above
26	Biometric and OCR	FR-BIO-001...009	side-service endpoints	face dialogs	enrolment store

#	Capability	Requirements	Principal API surface	Web route	Primary data
27	Access control	FR-RBAC-001...011	all endpoints	navigation and guards	roles, permissions, role_permissions, user_roles

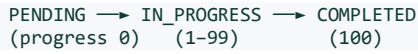
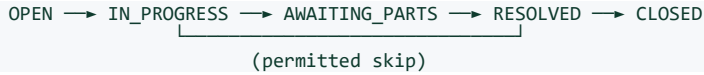
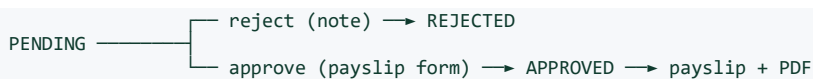
37. Appendices

Appendix A — Status Enumerations

Entity	States
Employee profile	PENDING, ACTIVE, INACTIVE, EXITED, OFFBOARDED
Attendance	PRESENT, ABSENT, WFH, HOLIDAY, LEAVE, HALF_DAY
Attendance mode	OFFICE, WFH, SITE, BIOMETRIC
Leave request	PENDING, APPROVED, REJECTED, CANCELLED
Leave accrual	ANNUAL, MONTHLY, MANUAL
Permission request	PENDING, APPROVED, REJECTED
Payroll run	PREVIEW, CONFIRMED, FINANCE_APPROVED, PAID
Payslip request	PENDING, APPROVED, REJECTED
Task	PENDING, IN_PROGRESS, COMPLETED
Task priority	LOW, MEDIUM, HIGH
Claim	PENDING, APPROVED, REJECTED
Asset	IN_STOCK, ASSIGNED, OUT_OF_STOCK, UNDER_REPAIR, RETIRED, LOST, DEPLOYED, BREAKDOWN
Asset return condition	GOOD, DAMAGED, LOST
Ticket	OPEN, IN_PROGRESS, AWAITING_PARTS, RESOLVED, CLOSED
Ticket priority	LOW, MEDIUM, HIGH, CRITICAL
Complaint / need kind	COMPLAINT, NEED
Complaint / need	OPEN, IN_REVIEW, RESOLVED, REJECTED
Safety incident type	NEAR_MISS, MINOR_INJURY, MAJOR_INJURY, PROPERTY_DAMAGE, ENV_HAZARD
Safety severity	LOW, MEDIUM, HIGH, CRITICAL
Safety incident	OPEN, INVESTIGATING, RESOLVED, CLOSED
Onboarding checklist	IN_PROGRESS, COMPLETED
Performance goal	ACTIVE, COMPLETED
Performance review	DRAFT, SUBMITTED, COMPLETED
Employment type	PERMANENT, CONTRACTUAL, DAILY_WAGE, SUBCONTRACTOR
Division	IT (displayed Digital), CIVIL (displayed Infra)

Appendix B — State Machines

Leave request

**Task — forward only****Support ticket — forward only, one step, with one permitted skip****Payroll run****Payslip request****Appendix C — Generated Identifier Formats**

Identifier	Format	Derivation
Employee code	EMP0001	Highest existing code with the prefix, plus one, four digits
Asset code	AST-2026-00001, INF-..., MCH-...	Category prefix, current year, five-digit sequence
Ticket code	TKT-2026-00001	Fixed prefix, current year, five-digit sequence
Complaint / need code	CN-2026-00001	Highest existing code for the year, plus one, five digits
Safety incident code	SI-2026-00001	Fixed prefix, current year, five-digit sequence
Refresh token	Two concatenated random UUIDs	Random
Stored file path	<folder>/<yyyy-MM>/<uuid>.<ext>	Folder, upload month, random name, original extension

Appendix D — Configuration Defaults

Setting	Default
Access token lifetime	4 hours
Refresh token lifetime	4 hours
Maximum failed sign-in attempts	5
Account lock duration	15 minutes
Default geofence radius	200 metres
Lateness grace	0 minutes
Office start / end	09:00 / 18:00
Standard work hours	8
Maximum upload per file / per request	25 MB / 60 MB

Setting	Default
Default page size	20
Database pool maximum / minimum idle	10 / 2
ESI wage ceiling / employee rate	21 000 / 0.75 %
Overtime hourly divisor	240
Ticket SLA — critical / high / medium / other	4 h / 8 h / 24 h / 48 h
Celebration horizon / list limit	60 days / 12 entries
Assistant history / knowledge context / ingestion cap	8 turns / 6 000 chars / 12 000 chars
Daily celebration job	09:00 server time
Server time zone	Asia/Kolkata
Default listen port	7060

Appendix E — Notification Catalogue

Trigger	Recipient	Type	Link
Leave applied	Named approver, or every leave approver	LEAVE	/leave/approvals
Leave decided	Applicant	LEAVE	/leave
Leave cancelled	Handling approver	LEAVE	/leave/approvals
Permission requested	Named approver	PERMISSION	/leave/permissions
Permission decided	Requester	PERMISSION	/leave/permissions
Payslip requested	Every payroll runner	PAYROLL	/payroll/requests
Payslip approved or rejected	Requester	PAYROLL	/payslips
Payroll run finance-approved	Every employee in the run	PAYROLL	/payslips
Task assigned, updated, progressed or completed	Assignee or assigner	TASK	/tasks
Claim submitted or updated	Approvers, or the owner	CLAIM	/ta-expenses
Claim decided	Owner	CLAIM	/ta-expenses
Asset allocated	Assignee	ASSET	/assets
Ticket raised, updated, reassigned, commented or progressed	Recipient, previous recipient, or raiser	HELPDESK	/helpdesk
Complaint or need submitted	Named recipient, or every HR and administrator	COMPLAINT	/complaints
Complaint or need updated	Submitter	COMPLAINT	/complaints
Safety incident reported	Every report viewer	SAFETY	/safety-incidents
Safety incident updated	Reporter	SAFETY	/safety-incidents
Holiday added	Every active employee	CALENDAR	/calendar
Birthday or work anniversary today	Every active employee except the celebrant	CELEBRATION	/
New chat message	Channel members	CHAT	/chat

Appendix F — SMS Catalogue

SMS is sent, in addition to the in-app notification, for: a leave application to its approver; a leave decision to the applicant; a permission request to its approver and its decision to the requester; a payslip request to payroll runners and its outcome to the requester; a task assignment, update and completion; a claim submission to approvers, a correction to the owner, and a decision to the owner; an asset allocation to the assignee; a ticket raised, commented and progressed; a complaint or need submitted and updated; a safety incident reported and updated; a new holiday, to the whole workforce; and the daily celebration, to the whole workforce except the celebrant.

Every SMS is prefixed with the organisation identifier, is dispatched asynchronously, and never blocks or fails the action that raised it.

Appendix G — Requirement Count Summary

Category	Count
Functional requirements	340
Non-functional requirements	92
Business rules	80
Data requirements	25
Interface requirements	34
Operational requirements	20
Constraints	15
Assumptions	8
Known limitations	15

Appendix H — Document Revision History

Version	Date	Change
1.0	30 July 2026	Baseline specification derived from the implemented system: full functional decomposition across 26 capability areas, access-control model, data requirements, consolidated business rules, non-functional requirements, deployment and operations, traceability and appendices.

End of document.